

Distribution robust water-based demand side management in power transmission networks

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ABSTRACT

The increasing share of renewable energy sources in the energy mix is posing new challenges for operating the power system. As a result, system operators must use the available flexibility more effectively to accommodate the surge in renewable energy. There are different ways to source flexibility including demand–response measures and sector integration. This paper investigates the utilization of flexibility from an integrated water and power network. A novel, augmented, distribution robust chance-constrained market clearing problem is proposed, assuming a centralized and fully competitive market set-up. Power generators and water pumps are modelled as flexible assets to procure flexibility from the integrated water and power network. Affine response policies and availability costs are defined to regulate the flexibility dispatch mechanism. The resulting coordination mechanism is nonlinear and non-convex. To make the problem feasible, different approximation and relaxation techniques are applied. A novel, hybrid, convex hull-based relaxation technique is proposed to convexify the friction head losses and pump energy consumption. The resulting problem is a second-order cone program which enables the power system operator to properly estimate and utilize the demand-side flexibility available in the water network. The results show that the electricity market benefits from grid flexibility obtained through the coordinated operation of controllable power generators and water pump loads. Under studied conditions, it was found that water pumps, when functioning as demand–response providers, can effectively compensate for power system imbalances, covering up to 25% of the total wind power mismatch. The sensitivity analysis emphasized the substantial impact of varying availability costs on the responses of flexible generators and water pumps. The error analysis identified the convex approximation of friction head losses as the primary source of inaccuracy in our model.

1. Introduction

1.1. Background and motivation

The increasing penetration of renewable energy sources (RES) in the energy mix is posing new challenges in operating the power network [1,2]. Renewable energy sources, such as wind and solar, are variable, partially unpredictable, and location-dependent. Therefore, large-scale integration of RES into the power network makes it increasingly difficult to control the power flows and to guarantee the stability and reliability of power networks [3]. As a result, system operators (SOs) must use the available flexibility more effectively to accommodate the surge in renewable energy [4]. Increased flexibility in the power sector can help to deal with uncertainty in a cost-efficient manner. There are different ways to source flexibility, such as demand–response (DR) services [5,6] and sector integration [7]. DR implies utilizing the flexibility that is available at the demand side, that is

exploiting the flexibility of end-users that are capable of shifting their consumption during peak periods [8]. When the flexibility available in one sector is not sufficient to ensure a reliable operation of the system, additional flexibility can be sourced from other sectors. This is commonly referred to as sector integration. Sector integration implies harnessing the flexibility that is available in one sector (e.g., water) to offer it to another sector (e.g., electricity) to handle the increasing level of uncertainty in the system [7]. Both DR services and sector integration have been shown to create new opportunities to produce, use and store energy more efficiently, thus mitigating the effects of increased uncertainty in the whole energy supply chain [7,8]. In this context, the integrated coordination of water and power networks could be a valuable option to procure flexibility from the water network to offer it to the power network. Such an integrated coordination mechanism, which is still largely missing in our societal resource management and infrastructure, could unveil significant operational and economic

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Nomenclature**Abbreviations**

<i>CC</i>	chance-constraint
<i>CP</i>	convex program
<i>DA</i>	day-ahead
<i>DC – OPF</i>	DC optimal power flow
<i>DC</i>	direct current
<i>DR</i>	demand–response
<i>DRCC</i>	distribution robust chance-constraint
<i>IEEE</i>	institute of electrical and electronics engineers
<i>MAE</i>	mean absolute error
<i>MCP</i>	market-clearing price
<i>MILP</i>	mixed-integer linear program
<i>MISOCP</i>	mixed-integer second-order cone program
<i>MSSP</i>	multi-stage stochastic programming
<i>MW</i>	megawatt
<i>NLP</i>	nonlinear program
<i>NRMSE</i>	normalized root mean square error
<i>OWF</i>	optimal water flow
<i>PAT</i>	pump as turbine
<i>PEP</i>	probability efficient point
<i>PTN</i>	power transmission network
<i>RES</i>	renewable energy sources
<i>RT</i>	real-time
<i>SCADA</i>	supervisory control and data acquisition
<i>SO</i>	system operator
<i>SOC</i>	second-order cone
<i>SOCP</i>	second-order cone program
<i>WTN</i>	water transport network

Number sets

$\mathbb{R}$	real numbers
D	conventional power loads
$\mathcal{E}$	water pipes
$\mathcal{G}$	conventional generators
$\mathcal{I}$	water network nodes
$\mathcal{J}$	water network junctions
$\mathcal{L}$	power transmission lines
$\mathcal{N}$	power buses
$\mathcal{P}$	centrifugal variable speed pumps
$\mathcal{R}$	water reservoirs
$\mathcal{S}$	elevated water storage tanks
$\mathcal{T}$	number of time slots
$\mathcal{V}$	pressure reducing valves
$\mathcal{W}$	wind power generators
$\mathcal{Z}$	number of random variables
Π	ambiguity set

Symbols

$\bar{\mathbf{x}}_{i,j}, \underline{\mathbf{x}}_{i,j}, \hat{\mathbf{x}}_{i,j}$	fraction of actual/maximum/minimum to reference rotational speed of pump i, j
$\alpha_{e,t}, \beta_{u,t}$	flexibility provided by generator e and pump u at time t

$\Delta\omega$	vector or random wind forecast error
$\delta_{w,t}$	wind power forecast error of wind farm w at time t
ϵ	violation probability level
€	euro
$\gamma_{i,j,t}, \zeta_{i,j,t}$	auxiliary variables for SOC relaxation of friction head losses and pump energy consumption for pipe/pump (i, j) at time t
$\mathbb{E}_{\mathbb{P}}$	expectation operator under probability distribution $\mathbb{P}$
$\mathbb{P}$	worst-case distribution of uncertainty selected from the ambiguity set Π .
μ^{Π}	mean vector of wind forecast error
ϕ^T	transpose vector of ones
Σ^{Π}	covariance matrix of wind forecast error
Σ_t	the t -th diagonal sub-matrix of the covariance matrix Σ^{Π} in time period t
τ	length of time step t
$\theta_{n,t}$	voltage angle of bus n at time t
$\underline{f}_{n,m}, \bar{f}_{n,m}$	lower and upper flow limits through power transmission line (n, m)
$\underline{H}_s^{end}, \bar{H}_s^{end}$	lower and upper bounds for the volume of storage tank s at the end of the simulation period $t = end$
$\underline{P}_{l,t}^D, \bar{P}_{l,t}^D$	lower and upper bounds for conventional load l at time t
$\underline{P}_{e,t}^G, \bar{P}_{e,t}^G$	lower and upper bounds for conventional power generator e at time t
$\underline{P}_{w,t}^W, \bar{P}_{w,t}^W$	lower and upper bounds for wind power generator w at time t
$\underline{P}_{i,j}, \bar{P}_{i,j}$	upper and lower bounds for head gain of pump (i, j)
$\underline{Q}_{i,j}, \bar{Q}_{i,j}$	minimum and maximum flow through pipe/pump/valve (i, j) at time t
a, b, b', c	hydraulic parameters of pump (i, j)
A_s	area of storage tank s
$B_{n,m}$	susceptance of power transmission line (n, m)
C_e^G	marginal production cost of generator e
C_e^{Av}, C_u^{Av}	availability cost of generator e and pump u
E_0	base voltage of the system
$F_{i,j}$	fixed part of Darcy–Weisbach equation for pipe (i, j) (flow-independent)
$g, \rho, \eta_{i,j}$	gravitational acceleration, density of water, and efficiency of pump (i, j)
$h_{j,t}, h_{r,t}, h_{s,t}$	pressure head at junction j , reservoir r and storage tank s at time t
$K_{i,j,t}$	pressure head change imposed by pressure reducing valve (i, j) at time t
$l_{i,j}^0, l_{i,j}^1$	parameters defining the convex envelope of the energy consumption curve for pump (i, j)
$m_{i,j,t}$	auxiliary variable for the McCormick relaxation of bilinear term (i, j) at time t

benefits, which can be realized in terms of lower operational costs and higher social welfare. Though important for practice and science, the study of such a coordination mechanism remains an ongoing area

of research, and the available literature on this topic is still limited. This work focuses on developing a new modelling approach resulting in an improved understanding of such an integrated energy and water coordination mechanism.

$p_{l,t}^D$	power demand of conventional load l at time t
$p_{e,t}^G$	power produced by generator e at time t
$p_{u,t}^P$	power demand of pump load u at time t
$P_{i,j,t}$	energy consumption of pump (i, j) at time t
$q_{i,j,t}^L, q_{i,j,t}^U$	lower and upper water flow bounds for the McCormick envelope
$r_{i,j}^0, r_{i,j}^1$	parameters defining the convex envelope of the friction head losses for pipe (i, j)
U_l^D	bid of conventional load l
U_u^P	bid of pump load u
$x_{i,j}, y_{i,j}, z_{i,j}$	empirical parameters describing the energy consumption of pump (i, j)
$d_{j,t}$	water demand at junction j at time t
$f_{n,m,t}$	power flow from bus n to bus m at time t
$H_j^0, \underline{H}_j, \overline{H}_j$	elevation, minimum and maximum pressure heads at junction j
$H_s^0, h_s^0, \underline{H}_s, \overline{H}_s$	elevation, initial water level, minimum and maximum pressure heads at storage tank s
$p_{u,t}^P$	power load of water pump u at time t
$p_{w,t}^W$	dispatch of wind power generator w at time t
$q_{i,j,t}$	water flow from node i to node j at time t
$V_{s,t}$	volume of water in storage tank s at time t

1.2. Literature review

In the domain of power systems research, various market-based coordination mechanisms have been introduced to investigate the planning and operation of integrated multi-carrier energy networks. These networks include diverse combinations such as power and gas networks [9,10], power and heat networks [11,12], and integrated power, heat, and hydrogen networks [13]. Such mechanisms have unveiled potential economic benefits and are considered valuable tools for business and strategic decision-making. However, little research has been conducted to develop a market-based coordination mechanism for integrated power and water networks. Traditionally seen as two independent streams and still widely managed separately, the link and strong interdependence of water and energy is nowadays acknowledged in the scientific and governmental fields [14,15]. This is generally referred to as the water-energy nexus [14–18]. Water is used for a variety of processes in energy production and, simultaneously, energy is used for water transport, treatment, and distribution. Water use in the energy sector accounts for approximately 10%–15% of total water withdrawals [19,20]. This percentage is mainly attributable to the cooling water needed for thermal power plants [21]. The water sector consumes 4%–7% of the electricity produced globally [22], of which 70%–80% is used for pumping (i.e., freshwater and groundwater withdrawal and transport), and 20%–30% is used for water treatment [23]. As power and water networks are physically and technologically interconnected, there are synergies across the two sectors that could be exploited to obtain operational flexibility. One way to study these synergies is to design integrated, market-based DR mechanisms which consider flexible loads in the water network. In a future DR program, DR operators could engage in contracts with the water network operator based on the flexibility they can offer. Therefore, DR operators need to foresee the operation of the power and water network in the near future, i.e. in the day-ahead (DA) stage, and estimate the potential flexibility they can trade. Evaluating the real potential of the DR at the system level is a difficult task which can be supported by mathematical models.

A fundamental challenge in developing a mathematical model for integrated, market-based power and water coordination mechanisms

lies in the structural difference between water and power systems and the associated decision-making mechanisms. Another challenge lies in modelling the nonlinear physics of power and water flows. Considering the water network only, several optimal water flow (OWF) problems have been proposed in the literature [24–26]. Refs. [24, 25] developed a mixed-integer second-order cone program (MISOCP) for energy-optimal pump scheduling and water flow. Such a mechanism delivers optimal pump schedules that minimize the cost of pump energy. Ref. [26] proposed a more comprehensive framework for optimizing the participation of water distribution network operators in DR and frequency regulation markets, taking into account the underlying hydraulic operating constraints. Their results reflected significant economic opportunities for the water sector when providing flexibility services in the market. However, all of the above-mentioned coordination mechanisms assumed fixed electricity pricing schemes and did not consider the dynamics of the power network. These approaches do not capture the impacts that the operation of one network (i.e., the water network) has on the other (i.e., the power network). In response, Ref. [27] developed an optimal water–power flow problem to optimize the use of controllable assets across power and water networks and proposed a successive convex approximation algorithm to identify feasible and optimal solutions. Ref. [28] proposed a quasi-convex hull relaxation technique to relax the optimal water–power flow problem and developed a coordination mechanism for demand-side management, showing that grid flexibility may indeed benefit from controllable water pump loads. However, the above-mentioned coordination mechanisms neglect the study of uncertainties inherent in the power and water networks, such as the uncertainties associated with power and water demands as well as renewable energy generation.

In a DR mechanism, it is particularly important to consider uncertainties, such as the variability of loads and RES generation, because their forecasts do not necessarily reflect their actual realization in real time (RT). Ref. [29] proposed a chance-constrained power and water coordination mechanism that considers pumps as flexible loads in the power distribution network. In their work, they consider uncertainties on the demand side (i.e., power loads and water demands) while they neglect uncertainties in renewable energy generation. In [29] the probabilistic chance constraints are approximated using the scenario-based stochastic programming approach. However, this approach is computationally intensive due to the large number of scenario simulations needed to accurately describe the probability distribution. Further, the scenario-based approach requires specific assumptions on the probability distribution of the stochastic variables. As discussed in [30,31], errors resulting from imperfect forecasting algorithms do not necessarily follow a specific probability distribution. In such cases, the distribution robust chance-constrained (DRCC) approach offers several advantages compared to other stochastic programming methods because it does not require any assumption on the probability distribution. Other advantages of the DRCC approach are that it is less conservative and more computationally efficient compared to other stochastic programming methods, such as scenario-based stochastic programming and robust optimization [32]. Most relevant to this study, Ref. [33] proposed a mixed-integer linear program (MILP) for the coordinated operation of water and electricity distribution networks considering variable renewable energy generation. The stochastic renewable energy generation is modelled using a data-driven probability efficient point (PEP) method based on historical wind and solar data. Head losses, pump hydraulics, and pump energy consumption are approximated with a piecewise linear function. Numerical results show the advantages of combining power and water networks and utilizing water pumps and tanks jointly as flexible loads. However, their approach considers the operational (i.e., real-time) stage only and neglects the DA scheduling stage. Considering only the operational stage may have a limiting effect on the amount of flexibility that can effectively be exchanged from the water sector to the power sector.

Table 1

Comparison of existing literature with current study considering network type, problem type, uncertainty modelling approach, and electricity market type.

Reference	Network type	Problem type	Uncertainty model	Market type
[6]	Power	SOCP	–	Balancing
[9]	Power/Gas	MILP	MSSP	DA & Balancing
[10]	Power/Gas	SOCP	DRCC	–
[24]	Water	MISOCP	–	–
[25]	Water	MISOCP	–	–
[26]	Water	MILP	–	Balancing
[27]	Power/Water	NLP	–	–
[28]	Power/Water	MICP	–	Balancing
[29]	Power/Water	CP	CC	Balancing
[33]	Power/Water	MILP	PEP	Balancing
This study	Power/Water	SOCP	DRCC	DA & Balancing

Further, a MILP with piecewise linear approximations is more computationally intensive compared to other methods, such as second-order cone programs (SOCPs), as discussed in [34]. What is missing, is a comprehensive market-based model that includes both DA scheduling and RT operation stages while accounting for the physical characteristics of both water and power networks and the technologies linking the two sectors. With regard to uncertainty modelling, more computationally efficient and generalized models and algorithms are still needed. A comparative analysis between the research presented in this paper and existing literature is detailed in Table 1.

1.3. Research aim

The research aims to develop an integrated, market-based electricity and water coordination mechanism to study the short-term scheduling and operation of integrated power and water networks taking into account the flexibility available in the water network. The proposed coordination mechanism is a SOC program that models the flexibility available at the supply side in the power network (i.e., flexible generators) and the demand side embedded in the water network (i.e., flexible operation of pumps and tanks). The utilization of demand-side flexibility embedded in the water network, which is the key element of this work, is also referred to as water-based demand-side management. The proposed model is formulated as an augmented DA market clearing problem that maximizes utility in the power sector while retaining enough flexibility in the system to compensate for possible power imbalances that could be realized at a later stage (i.e., the RT stage). The proposed mathematical program can also be interpreted as a joint DA and RT optimal power–water flow problem with power generators acting as flexible producers and water pumps acting as flexible consumers in the power network. The feasible operation of the water pumps is limited by water network constraints. Wind power forecasts are considered the only source of uncertainty in the system. Considering the advantages over other stochastic programming approaches, the problem is formulated as a data-driven DRCC program. The problem is robust as it considers the worst-case probability distribution among a set of possible probability distributions, also known as an ambiguity set, defined based on available historical data of wind power forecasts. Convex approximations and affine control policies are implemented to make the DRCC program computationally tractable. To the best of the author's knowledge, a DRCC program has never been developed for the short-term scheduling and operation of integrated power and water networks while considering uncertainty in renewable energy generation. The novelty of this work lies in the formulation and algorithmic implementation of a DRCC mathematical programming problem in the context of integrated power and water networks. A novel relaxation technique is proposed to convexify the friction head losses and pump energy consumption functions. The proposed DRCC framework is unique in that it helps to identify synergies across interdependent power and water networks and proposes a market participation strategy for water pumps acting as flexible loads in the power network. The authors believe that this contribution represents a step forward in understanding the potential benefits and drawbacks of integrating the power and water sectors.

1.4. Contributions

This paper proposes a novel water network-aware electricity market-clearing mechanism that accounts for the physically feasible operation of the water distribution network (e.g., water network-aware clearing). The problem is formulated as a DRCC mathematical optimization problem that seeks to maximize utility in the power sector while retaining enough operational flexibility in the system to compensate for possible power imbalances that could be realized at a later stage. This is also referred to as an augmented, regret-free market clearing mechanism. Assuming a centralized and fully competitive market setup, the optimization model includes physical constraints of the water network (including pipe head loss and pump pressure head gain), and of the power network (e.g., power flows and line capacities). Affine policies and availability costs are defined to regulate the response of flexible agents in the integrated power and water system (i.e., flexible power generators and pump loads). Fig. 1 provides a detailed schematic overview of the proposed coordination mechanism. It outlines the objective function (top), the primary physical processes, and their corresponding decision variables for the DA stage (middle), and the decision variables related to the allocation of flexibility reserves in the RT stage (bottom). The continuous arrows represent the model's output concerning the DA stage, whereas the dashed arrows depict the model's flexibility response in RT.

The resulting optimization model is nonlinear and non-convex. To make the problem feasible, different relaxation and approximation techniques are adopted to convexify the water and power flows. A novel, hybrid, convex hull-based relaxation method is proposed to convexify the friction head losses and electric pump models. The DRCCs are approximated using the SOC relaxation method based on a variant of Chebyshev's Inequality [35,36]. The applicability of the proposed approach is demonstrated with a case study.

The contributions of the paper are the following:

1. proposes a novel, market-based, DRCC demand–response framework to study flexibility provision from the water network to the power network considering uncertainty in renewable energy generation.
2. formulates the framework as a joint DA and RT mathematical optimization problem that seeks to maximize utility in the power sector while retaining enough flexibility in the system to compensate for possible power imbalances in real time.
3. proposes a novel, hybrid, convex hull-based relaxation technique to convexify the Darcy–Weisbach equation and pump energy consumption.
4. shows the sensitivity of the DR mechanism to the availability costs of flexible assets.

1.5. Paper outline

The remainder of this paper is organized as follows. Section 2 presents the formulation of the DRCC augmented market clearing problem, including flexible power generation and water-based demand-side

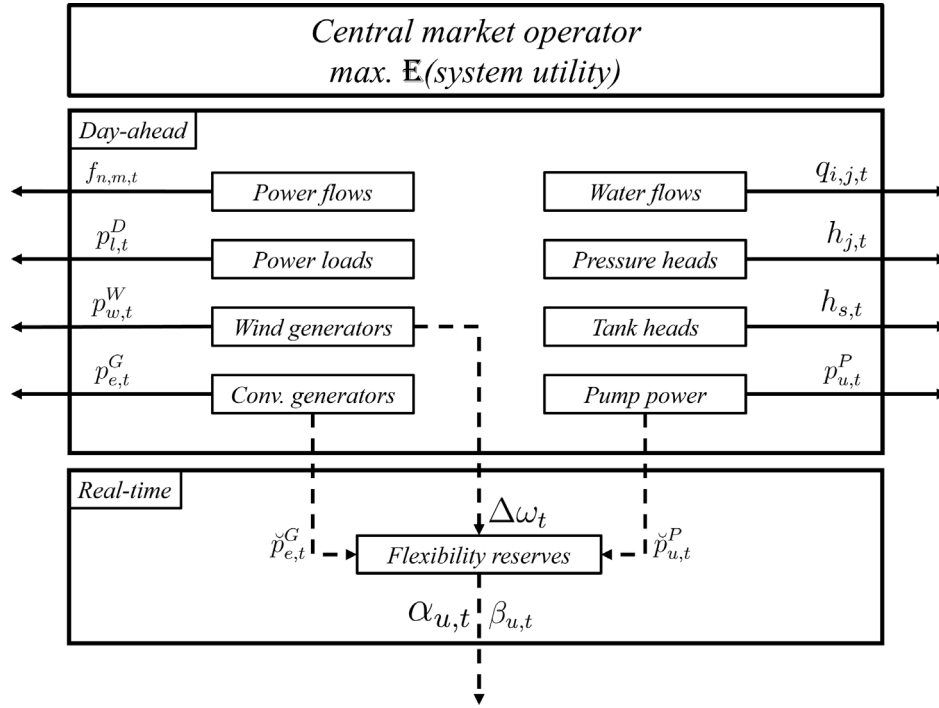


Fig. 1. Schematic overview of the proposed augmented DA market clearing mechanism including nominal DA decisions (continuous line), and uncertainty-dependent RT decisions (dashed line).

management. Section 3 discusses the solution methodology, including the description of uncertainty propagation as well as different approximations and relaxations to make the problem tractable. The applicability of the proposed method is demonstrated throughout a case study in Section 4. Finally, conclusions are drawn and points for future research are highlighted in Section 5.

2. Problem formulation

2.1. Modelling assumptions

Electric water pumps are assumed to be the major technologies that link the power and water networks. The reason is that the electricity consumption associated with pumping can represent up to 80% of the total energy consumption in the water sector. As per flexibility, it is assumed that water pumps coupled with storage tanks can serve as a flexible load type in the power network. Additionally, generators in the power network can also function as flexible assets by increasing or decreasing power production within their operational limits. On the power side, a single-area Power Transmission Network (PTN) is considered. The PTN includes a set of transmission lines $\mathcal{L}$ and a set of nodes $\mathcal{N}$. Each node may contain uncontrollable loads $l \in \mathcal{D}$, controllable pump loads $u \in \mathcal{P}$, conventional power generating units $e \in \mathcal{G}$, and wind generators $w \in \mathcal{W}$. In this study, we consider generic flexible power generators representative of real-world scenarios, such as gas-fired power plants. To model the power transmission network, a linearized DC optimal power flow (DC-OPF) formulation is adopted. In the linear DC-OPF formulation, voltage magnitudes are assumed to be close to 1 per unit. Power losses are neglected. Further, resistance is assumed to be small compared to reactance. As voltage angle differences over the network elements are assumed to be small, the linear approximation can be used to model the sine function. Since this study focuses on the power transmission level and the main objective is to study the integration of the water network with the power network, the linearized DC-OPF formulation is considered sufficiently accurate. A more detailed description of power flows applicable to power distribution networks is left for future work.

On the water side, a pressurized water transport network is considered (WTN). The WTN is described as a graph consisting of several nodes that can be categorized as water reservoirs, water storage tanks, or junctions. The different nodes are connected with a set of water pipes carrying a certain flow of water through the system. The system relies on centrifugal variable speed pumps (VSPs) to overcome the pressure head differences due to the different elevations at different nodes, as well as the friction head losses occurring in the pipes. The pump status (i.e., on/off) is assumed to be determined in advance of the DA scheduling horizon. Further, it is assumed that water flow directions are predetermined and do not change. This enables us to eliminate the need for binary variables which would lead to a more computationally intensive mixed-integer problem as in [37,38]. These assumptions are commonly adopted in water network modelling [24,29]. Following the approach from [29,33], the water network is described as a directed graph consisting of a set of nodes $\mathcal{I}$ and a set of edges $\mathcal{E}$. Nodes may be defined as junctions $j \in \mathcal{J}$, reservoirs $r \in \mathcal{R}$, or elevated storage tanks $s \in \mathcal{S}$, such that $\mathcal{I} = \mathcal{J} \cup \mathcal{R} \cup \mathcal{S}$. The difference between reservoirs and tanks is that reservoirs are modelled as an infinite water source, while tanks have a specific dimension and therefore a finite water volume. Further, it is assumed that reservoirs are located at a lower elevation compared to storage tanks so that water can be pumped into elevated storage tanks at certain times and then be released through gravitational potential energy at later periods. Edges are water pipes which can be further categorized as supply pumps $(i, j) \in \mathcal{P}$ or as pressure-reducing valves $(i, j) \in \mathcal{V}$, such that $(\mathcal{P} \cup \mathcal{V}) \subset \mathcal{E}$.

It is assumed that both power and water systems are scheduled with a time resolution of one hour. It must be noted that the two systems are not necessarily scheduled and operated with the same time resolution. The decisions in the water system are usually readjusted at rougher granularity compared with the power system (e.g., daily vs. hourly). However, the assumption that the two systems are scheduled with the same time resolution (e.g., hourly) makes it possible to study a complex decision-making dynamic. This is common practice in both power and water systems scheduling [24]. Further, it has been shown that, with the development of new sensor technologies and supervisory control and data acquisition (SCADA) systems, it is possible

to implement RT control of the water network working at very high time resolutions [29]. Without loss of generality, the assumption can easily be relaxed, by assuming that the daily water schedule is the same throughout the day. Therefore, regardless of the granularity at which decisions in water and power systems are made, the modelling approach proposed is entirely general.

Wind power generation is considered to be the only source of uncertainty in the model. Although our uncertainty modelling approach is completely general, our focus is on creating a tractable DRCC formulation for integrated power and water networks. Exploring other uncertainties, such as solar power generation, power loads and water demands, is beyond the scope of this work and is left for future research. Wind power uncertainty is modelled with a moment-based ambiguity set which describes the uncertainty of DA wind power forecast errors. Following the approach from [10,31], it is assumed that wind power forecast errors follow an unknown multivariate probability distribution which is continuously differentiable. The overall system uncertainty is described by a vector of random variables $\Delta\omega = [\delta_{1,1}, \delta_{2,1}, \dots, \delta_{w,t}] \in \mathbb{R}^Z$. Consequently, the moment-based ambiguity set embracing all probability distributions described by the same first and second-order moments can be represented as follows

$$\Pi = \{\mathbb{P} \in \Pi_0(\mathbb{R}^Z) : \mathbb{E}_{\mathbb{P}}[\Delta\omega] = \mu^\Pi, \mathbb{E}_{\mathbb{P}}[\Delta\omega\Delta\omega^T] = \Sigma^\Pi\},$$

which implies that the set of plausible probability distributions $\Pi_0(\mathbb{R}^Z)$ contains all probability distributions that share the same first and second-order moments of wind forecast errors, i.e. mean and covariance, given by known parameters $\mu^\Pi \in \mathbb{R}^Z$ and $\Sigma^\Pi \in \mathbb{R}^{Z \times Z}$ respectively. The expectation with respect to the probability distribution is denoted by $\mathbb{E}_{\mathbb{P}}$, and superscript T is the transpose operator. Without loss of generality, it is assumed that the mean $\mu^\Pi = 0$ and that the covariance matrix Σ^Π can appropriately be estimated from empirical data of wind forecast errors. The covariance matrix Σ^Π is positive semi-definite and its diagonal entries describe the spatial relationship between wind forecast errors at time t (i.e., the correlation of wind forecast errors between zone 1 and zone 2 at time t), while the off-diagonal entries describe the temporal dependence of the uncertain parameters (i.e., the correlation of wind forecast errors between time $t-1$ and time t) as explained in [39]. With this description of uncertainty, the total deviation from the DA forecast of all wind farms at time t is given by $\phi^T \Delta\omega_t$, where $\phi^T \in \mathbb{R}^Z$ is a vector of all ones. Here, $\phi^T \Delta\omega_t \geq 0$ implies a shortage of wind power available during RT operation compared to the DA point forecast. In this study, we focus on the case of wind power deficit. The reason is that, from the system operator's perspective, the worst-case situation occurs during wind deficits. While excess wind power can be curtailed, a wind deficit necessitates compensation using the flexibility available in the system.

2.2. Day-ahead power and water coordination

The proposed DA power and water coordination mechanism is a DRCC program, presented in (1)–(3) as follows. The objective function (1a) has a min–max structure such that the total social disutility of the power system is minimized (or utility is maximized) while uncertainty is drawn from the worst-case probability distribution, which is the probability distribution that results in the maximum expected social disutility due to uncertainty realization.

$$\underset{\Gamma, \mathbb{P} \in \Pi}{\text{MinMax}} \quad \mathbb{E}_{\mathbb{P}} \left[\sum_{i \in \mathcal{T}} \left(\sum_{e \in \mathcal{G}} C_e^G(\bar{p}_{e,i}^G) - \sum_{l \in \mathcal{D}} U_l^D \bar{p}_{l,i}^D - \sum_{u \in \mathcal{P}} U_u^P \bar{p}_{u,i}^P \right) \right] \quad (1a)$$

s.t.

Power constraints

$$\bar{p}_{l,i}^D \leq p_{l,i}^D \leq \bar{p}_{l,i}^D \quad l \in \mathcal{D}, t \in \mathcal{T} \quad (1b)$$

$$\bar{p}_{w,i}^W \leq \bar{p}_{w,i}^W \leq \bar{p}_{w,i}^W \quad w \in \mathcal{W}, t \in \mathcal{T} \quad (1c)$$

$$\underset{\mathbb{P} \in \Pi}{\text{min}} \mathbb{P}[\bar{p}_{e,i}^G \leq \bar{p}_{e,i}^G \leq \bar{p}_{e,i}^G] \geq (1 - \epsilon) \quad e \in \mathcal{G}, t \in \mathcal{T} \quad (1d)$$

$$\sum_{e \in \mathcal{G}_n} \bar{p}_{e,i}^G + \sum_{u \in \mathcal{P}_{n_n}} p_{u,i}^W = \sum_{l \in \mathcal{D}_n} p_{l,i}^D + \sum_{u \in \mathcal{P}_{n_n}} \bar{p}_{u,i}^P + \sum_{m \in \Omega_n} \bar{f}_{n,m,i} \quad n \in \mathcal{N}, t \in \mathcal{T} \quad (1e)$$

$$\underset{\mathbb{P} \in \Pi}{\text{min}} \mathbb{P}[\bar{f}_{n,m} \leq \bar{f}_{n,m} \leq \bar{f}_{n,m}] \geq (1 - \epsilon) \quad (n, m) \in \mathcal{L}, t \in \mathcal{T} \quad (1f)$$

$$\bar{f}_{n,m,i} = E_0^2 B_{n,m}(\bar{\theta}_{n,i} - \bar{\theta}_{m,i}) \quad (n, m) \in \mathcal{L}, t \in \mathcal{T} \quad (1g)$$

$$\bar{\theta}_{n,i} = 0 \quad n = ref, t \in \mathcal{T} \quad (1h)$$

Water constraints

$$\underset{\mathbb{P} \in \Pi}{\text{min}} \mathbb{P}[\bar{H}_j + H_j^0 \leq \bar{h}_{j,t} \leq \bar{H}_j + H_j^0] \geq (1 - \epsilon) \quad j \in \mathcal{J}, t \in \mathcal{T} \quad (2a)$$

$$h_{r,t} = H_{r,t}^0 \quad r \in \mathcal{R}, t \in \mathcal{T} \quad (2b)$$

$$\sum_{i \in \mathcal{N}_{in}^J} \bar{q}_{i,j,t} - \sum_{k \in \mathcal{N}_{out}^J} \bar{q}_{j,k,t} = d_{j,t} \quad j \in \mathcal{J}, t \in \mathcal{T} \quad (2c)$$

$$\bar{V}_{s,t} = V_s^0 + \tau \left(\sum_{i \in \mathcal{N}_{in}^S} \bar{q}_{i,s,t} - \sum_{j \in \mathcal{N}_{out}^S} \bar{q}_{s,j,t} \right) \quad s \in \mathcal{S}, t = 1 \quad (2d)$$

$$\bar{V}_{s,t} = \bar{V}_{s,t-1} + \tau \left(\sum_{i \in \mathcal{N}_{in}^S} \bar{q}_{i,s,t} - \sum_{j \in \mathcal{N}_{out}^S} \bar{q}_{s,j,t} \right) \quad s \in \mathcal{S}, t > 1 \quad (2e)$$

$$\underset{\mathbb{P} \in \Pi}{\text{min}} \mathbb{P}[\bar{V}_s \leq \bar{V}_{s,t} \leq \bar{V}_s] \geq (1 - \epsilon) \quad s \in \mathcal{S}, t \in \mathcal{T} \quad (2f)$$

$$\bar{h}_{s,t} - h_s^0 = \frac{\tau}{A_s} \left(\sum_{i \in \mathcal{N}_{in}^S} \bar{q}_{i,s,t} - \sum_{j \in \mathcal{N}_{out}^S} \bar{q}_{s,j,t} \right) \quad s \in \mathcal{S}, t = 1 \quad (2g)$$

$$\bar{h}_{s,t} - \bar{h}_{s,t-1} = \frac{\tau}{A_s} \left(\sum_{i \in \mathcal{N}_{in}^S} \bar{q}_{i,s,t} - \sum_{j \in \mathcal{N}_{out}^S} \bar{q}_{s,j,t} \right) \quad s \in \mathcal{S}, t > 1 \quad (2h)$$

$$\underset{\mathbb{P} \in \Pi}{\text{min}} \mathbb{P}[\bar{H}_s + H_s^0 \leq \bar{h}_{s,t} \leq \bar{H}_s + H_s^0] \geq (1 - \epsilon) \quad s \in \mathcal{S}, t \in \mathcal{T} \quad (2i)$$

$$\underset{\mathbb{P} \in \Pi}{\text{min}} \mathbb{P}[\bar{H}_s^{end} \leq \bar{h}_{s,t} \leq \bar{H}_s^{end}] \geq (1 - \epsilon) \quad s \in \mathcal{S}, t = end \quad (2j)$$

$$\bar{h}_{i,t} - \bar{h}_{j,t} = \bar{K}_{i,j,t} \quad (i, j) \in \mathcal{V}, t \in \mathcal{T} \quad (2k)$$

$$\underset{\mathbb{P} \in \Pi}{\text{min}} \mathbb{P}[\bar{Q}_{i,j} \leq \bar{q}_{i,j,t} \leq \bar{Q}_{i,j}] \geq (1 - \epsilon) \quad (i, j) \in \mathcal{V}, t \in \mathcal{T} \quad (2l)$$

$$\bar{h}_{i,t} - \bar{h}_{j,t} = F_{i,j}(\bar{q}_{i,j,t})^2 \quad (i, j) \in \mathcal{E}, t \in \mathcal{T} \quad (2m)$$

$$\underset{\mathbb{P} \in \Pi}{\text{min}} \mathbb{P}[\bar{Q}_{i,j} \leq \bar{q}_{i,j,t} \leq \bar{Q}_{i,j}] \geq (1 - \epsilon) \quad (i, j) \in \mathcal{E}, t \in \mathcal{T} \quad (2n)$$

$$\bar{h}_{j,t} - \bar{h}_{i,t} = a_{i,j} \bar{q}_{i,j,t}^2 + \mathfrak{N}_{i,j} b_{i,j} \bar{q}_{i,j,t} + \mathfrak{N}_{i,j}^c c_{i,j} \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (2o)$$

$$\underset{\mathbb{P} \in \Pi}{\text{min}} \mathbb{P}[\bar{Q}_{i,j} \leq \bar{q}_{i,j,t} \leq \bar{Q}_{i,j}] \geq (1 - \epsilon) \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (2p)$$

$$\bar{P}_{i,j,t} = \frac{g\rho\tau}{\eta_{i,j}} (\bar{h}_{j,t} - \bar{h}_{i,t}) \bar{q}_{i,j,t} \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (2q)$$

Coupling constraints

$$\bar{p}_{u,i}^P = \bar{P}_{i,j,t}, \quad u \in \mathcal{P}_{i,j}, (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (3a)$$

$\Gamma = \{p_{l,i}^D, \bar{p}_{w,i}^W, \bar{p}_{e,i}^G, \bar{p}_{u,i}^P, \bar{f}_{n,m,i}, \bar{\theta}_{n,i}, \bar{h}_{i,t}, h_{r,t}, \bar{q}_{i,j,t}, \bar{h}_{s,t}, \bar{V}_{s,t}, \bar{P}_{i,j,t}, \bar{K}_{i,j,t}\}$ is the set of optimization decision variables. The symbol tilde ($\sim$) means that the decision variable includes both its nominal and uncertainty-dependent parts. The separation of nominal and uncertainty-dependent parts and the modelling of uncertainty propagation is discussed in Sections 2.3 and 2.4. Objective (1a) minimizes the total expected disutility of the power system (or maximizes system utility). Parameters U_l^D , U_u^P and C_e^G represent the estimated bids of conventional load l , pump load u , and the marginal production cost of generator e respectively. Notably, the marginal production cost of generators is represented as a stepwise aggregated cost curve, aligning with prevailing energy market practices within the European context. The electricity demand is considered to be elastic, denoting the capacity of conventional loads to adjust their consumption in response to different market conditions. The power generation of wind producers is excluded from the objective function as a zero marginal wind power production cost is assumed. Further, it is assumed that water pump bids U_u^P are set at the maximum auction

price (i.e., price cap or the bid with the highest price) as pump loads must be fully dispatched at all times to ensure water supply–demand matching.

The inequalities (1d), (1f), (2a), (2f), (2i), (2j), (2l), (2n) and (2p) are modelled as DRCCs. This implies that at the optimal solution, the probability of meeting each constraint is modelled to have a violation probability of at most ϵ or, in other words, a confidence level of at least $(1 - \epsilon)$. Constraints (1b)–(1h) describe the power transmission network. These constraints include load and generation limits (1b)–(1d), the power balance (1e), the power flow limits through the transmission lines (1f), the power flow definition (1g), and the fixed voltage angle at the slack node (1h). The slack node is used to represent the interaction of the power network with the external nodes. It makes sense not to connect any loads or generators directly to the slack node, as it does not influence the power flow physics. Variables $\bar{p}_{e,t}^G$, $\bar{p}_{w,t}^W$, $\bar{p}_{l,t}^D$, and $\bar{p}_{u,t}^D$ represent the power produced by generator e , wind farm w , electricity demand from load l , and water pumps u in period t respectively. Variable $\bar{\theta}_{n,t}$ represents the voltage angle at bus n at time t . Variable $\bar{f}_{n,m,t}$ describes the power flow from bus n to bus m at time t . Parameters $\bar{p}_{l,t}^D$, $\underline{p}_{l,t}^D$, $\bar{p}_{w,t}^W$, $\underline{p}_{w,t}^W$, $\bar{p}_{e,t}^G$, $\underline{p}_{e,t}^G$ represent the upper and lower limits of loads, wind power generators, and conventional power generators respectively. Parameters $\bar{f}_{n,m}$, $\underline{f}_{n,m}$ are the maximum and minimum power flow limits for line n, m respectively. Parameter $B_{n,m}$ is the susceptance of the transmission line connecting node n to node m . Parameter E_0 is the base voltage magnitude which is assumed to be close to one per unit. It is conventional to assume that the voltage magnitude at each node is equal to the base voltage [40].

Constraints (2a)–(2q) describe the water network. These constraints include pressure head limits (2a), fixed head at the reservoir (2b), flow balance (2c), the evolution of tank volume and volume limits (2d)–(2f), the evolution of tank head and head limits (2g)–(2j), pressure head drop associated with pressure reducing valves and valve flow limits (2k)–(2l), pipe friction head loss and pipe flow limits (2m)–(2n), pump hydraulic head gain and flow limits (2o)–(2p), and pump energy consumption (2q). Variables $\bar{h}_{j,t}$, $\bar{h}_{r,t}$, $\bar{h}_{s,t}$ represent the pressure head at junction j , reservoir r , and storage tank s at time t respectively. Variable $\bar{q}_{i,j,t}$ represents the water flow from node i to node j at time t . Variables $\bar{v}_{s,t}$ and $\bar{h}_{s,t}$ represent the volume and head of storage tank s at time t . $\bar{\kappa}_{i,j,t}$ is an auxiliary variable describing the pressure head change imposed by pressure-reducing valve i, j at time t . Variable $\bar{p}_{i,j,t}$ represents the energy consumption of water pump i, j at time t . Parameters H_j^0 , $\underline{H}_j$, $\bar{H}_j$ represent the elevation, minimum, and maximum head at junction j . Parameter $d_{j,t}$ describes the water demand at junction j and time t . Parameters $\bar{v}_s$, $\underline{v}_s$ represent the maximum and minimum volume in tank s . Parameters $\bar{Q}_{i,j}$, $\underline{Q}_{i,j}$ represent the maximum and minimum flow through pipe i, j . Parameter $F_{i,j}$ represents the fixed part of the Darcy–Weisbach equation (i.e., the flow-independent part). Parameter τ is the length of the time step t . Parameter A_s is the cross-sectional area of the tank. In (2o), $a_{i,j}$, $b_{i,j}$, and $c_{i,j}$ are parameters describing the hydraulic behaviour of pump i, j ; $\mathbb{N}_{i,j}$ is the fraction of actual to reference speed of pump i, j which is used to correct the pump hydraulic curve based on the actual operating speed. In (2q) constants ρ , g , and $n_{i,j}$ represent the density of water, the gravitational acceleration, and the efficiency of pump i, j respectively.

It is important to note that constraint (2b) sets the pressure head at the reservoir equal to the elevation under the assumption that reservoirs are infinite water sources. Constraint (2j) sets an upper bound $\bar{H}_s^{end}$ and lower bound $\underline{H}_s^{end}$ on the final head (or level) of storage tank s at $t = end$. These constraints are enforced to ensure that storage tanks are not completely filled or emptied at the end of the simulation period. The reason is that a completely filled or emptied tank would affect the operation of the system on the next day and it would create a bias in the interpretation of the results. Thus, while tanks can be filled or emptied throughout the day, their level should remain within a predefined range at the end of the 24 h.

It must be noted that this is not a strong assumption, and therefore these constraints can easily be relaxed without loss of generality of the model. Further, constraint (2k) models pressure-reducing valves, where $\bar{\kappa}_{i,j,t}$ is a decision variable describing the pressure drop associated with valve i, j at time t . This is required to maintain a stable operation of the water network. Pressure-reducing valves are installed at every junction with two or more incoming pipelines. Finally, constraint (3a) couples the electricity consumption of water pumps with the respective loads on the power network, thus representing the link between the power and the water network. Note that program (1)–(3) enables the market operator to source flexibility that is at the disposal of the water network (i.e., flexibility that is embedded in the existing water network infrastructure). If not utilized, this flexibility would be completely lost. If utilized, it would not pose a risk to water consumers. This ensures that the water system is not jeopardizing the security of supply to the end-users by providing flexibility to the power network.

At this stage, the DRCC program (1)–(3) is computationally intractable. The reasons are manifold. First of all, program (1)–(3) contains nonlinear and nonconvex constraints (2m), (2o) and (2q). Secondly, DRCC optimization problems in their original form are difficult to solve because they include an infinite number of variables and an infinite number of constraints [41]. The DRCC program (1)–(3) is a worst-case expectation problem which contains probabilistic chance constraints that are functions of random variables following an unknown probability distribution. This leads to an optimization problem of infinite dimensions which is computationally intractable [10,31]. However, by leveraging affine control policies, convex approximations, and duality theory for moment problems, DRCC problems become computationally tractable across a wide variety of ambiguity sets [42].

To make the DRCC program (1)–(3) computationally tractable the following procedure is adopted. Firstly, linear decision rules are defined to model the response of flexible assets as affine functions of uncertainty (i.e., as a linear function of wind forecast errors). Linear decision rules provide a simplified and intuitive understanding of uncertainty propagation from the power network to the water network. These are control actions that enable the market operator to regulate flexible power generators and water pumps in the coupled power–water network. Secondly, different approximation and relaxation techniques are proposed to convexify constraints (2m), (2o) and (2q) in the program (1)–(3). Thirdly, the DRCCs are reformulated as SOC constraints based on a variant of Chebyshev's Inequality [35,36].

2.3. Affine uncertainty response policies

Affine response policies are adopted to regulate the amount of flexibility provided by flexible assets, namely conventional power generators and water pumps, in the coupled power–water network on the DA stage. That is, flexibility reserves are allocated on top of the nominal DA schedule in response to the worst-case expected wind power deviation in the RT stage. Let us define the total stochastic wind power generation $\bar{p}_{w,t}^W$ by two separate components: the DA wind power forecast $\bar{p}_{w,t}^W$ and the wind power deviation resulting from forecast errors, denoted as a vector of random variables $\Delta\omega_t = [\delta_{1,t}, \delta_{2,t}, \dots, \delta_{w,t}]$ such that $\bar{p}_{w,t}^W = \bar{p}_{w,t}^W + \delta_{w,t}$. The response of flexible assets in RT is therefore modelled as an affine function of the total aggregated wind power mismatch $\phi^T \Delta\omega_t$. Conventional power generators are capable of increasing or decreasing their production in response to wind forecast errors. The uncertainty response by power generators is modelled as

$$\bar{p}_{e,t}^G = \bar{p}_{e,t}^G + \bar{p}_{e,t}^G (\phi^T \Delta\omega_t) \quad e \in \mathcal{G}, t \in \mathcal{T} \quad (4)$$

where $\bar{p}_{e,t}^G$ is the total stochastic power generation of generator e at time t , which is composed of the nominal power generation $\bar{p}_{e,t}^G$ and the uncertainty-dependent generation response $\bar{p}_{e,t}^G$ denoting the participation factor of unit e towards compensating the total wind power mismatch $\phi^T \Delta\omega_t$ at time t .

Similarly, water pumps are capable of increasing or decreasing their power consumption in response to uncertainty realization. This is modelled as

$$\tilde{p}_{u,t}^P = p_{u,t}^P + \check{p}_{u,t}^P (\phi^T \Delta \omega_t) \quad u \in \mathcal{P}, t \in \mathcal{T} \quad (5)$$

where $\tilde{p}_{u,t}^P$ is the total stochastic power demand of pump u at time t , which is composed of the nominal power demand $p_{u,t}^P$ and the uncertainty-dependent demand-response $\check{p}_{u,t}^P$ denoting the participation factor of unit u towards compensating the total wind power mismatch $\phi^T \Delta \omega_t$ at time t . The effects of uncertainty responses by flexible assets, i.e., changes in power flows, water flows and water tank levels, are considered indirect as they are dependent on the allocation of the above recourse actions and subject to the physical constraints of the network. The following paragraph discusses how uncertainty propagates to the power and water networks.

2.4. Modelling uncertainty propagation

Considering the case of imperfect wind forecasts, i.e., $\phi^T \Delta \omega_t \neq 0$, the allocation of affine policies (4) and (5) above will inevitably affect the power and water balance constraints as well as the flows in the power and water networks. Uncertainty propagation on the power side affects the power balance constraint (1e) and the power flows in the transmission lines modelled in (1g). Given the affine response from power generators $\check{p}_{e,t}^G$ and water pump loads $\check{p}_{u,t}^P$, the power balance can be modelled by separating the nominal and uncertainty-dependent parts of (1e) as follows

$$\sum_{e \in \Psi_{Gn}} p_{e,t}^G + \sum_{w \in \Psi_{Wn}} p_{w,t}^W = \sum_{l \in \Psi_{Dn}} p_{l,t}^D + \sum_{u \in \Psi_{Pn}} p_{u,t}^P + \sum_{m \in \Omega n} f_{n,m,t} \quad n \in \mathcal{N}, t \in \mathcal{T} \quad (6a)$$

$$\sum_{e \in \mathcal{G}} \check{p}_{e,t}^G - \sum_{u \in \mathcal{P}} \check{p}_{u,t}^P = 1 \quad t \in \mathcal{T} \quad (6b)$$

$$\sum_{e \in \Psi_{Gn}} \check{p}_{e,t}^G - \sum_{u \in \Psi_{Pn}} \check{p}_{u,t}^P - \sum_{m \in \Omega n} \check{f}_{n,m,t} = 0 \quad n \in \mathcal{N}, t \in \mathcal{T} \quad (6c)$$

where (6a) is the nominal part of (1e) while (6b) and (6c) represent the uncertainty-dependent counterparts. Similarly, power flows change depending on the affine response of power generators $\check{p}_{e,t}^G$ and water pumps $\check{p}_{u,t}^P$. Therefore, the power flows can be modelled by separating the nominal and uncertainty-dependent part of (1g) and (1h) as follows

$$f_{n,m,t} = E_0^2 B_{n,m} (\theta_{n,t} - \theta_{m,t}) \quad (n, m) \in \mathcal{L}, t \in \mathcal{T} \quad (7a)$$

$$\check{f}_{n,m,t} = \phi^T \Delta \omega_t [E_0^2 B_{n,m} (\check{\theta}_{n,t} - \check{\theta}_{m,t})] \quad (n, m) \in \mathcal{L}, t \in \mathcal{T} \quad (7b)$$

$$\theta_{n,t} = 0 \quad n = ref, t \in \mathcal{T} \quad (7c)$$

$$\check{\theta}_{n,t} = 0 \quad n = ref, t \in \mathcal{T} \quad (7d)$$

where (7a) and (7c) are the nominal parts while (7b) and (7d) are the uncertainty-dependent counterparts of constraint (1g) and (1h) respectively. On the water side, the flexible operation of water pumps, which is determined by the affine response $\check{p}_{u,t}^P$, in turn, affects the water flows in the pipelines, the pressure heads at the different nodes, and the volume in elevated storage tanks. Here, it is important to realize that these changes should be modelled as a function of $\check{p}_{u,t}^P$. Yet, given that water flows, friction head losses and pump hydraulics are non-linear, the derivation of this relationship would make the problem more complex. Instead, adjustments in water flows, pressure head changes, friction head losses, and pump hydraulics are modelled as affine functions of uncertainty. For the purpose of this work, this approach is considered adequate to capture the required physics of the system. Water flows in the pipelines are therefore modelled as

$$\tilde{q}_{n,m,t} = q_{n,m,t} + (\phi^T \Delta \omega_t) \check{q}_{n,m,t} \quad (n, m) \in (\mathcal{E} \cup \mathcal{V} \cup \mathcal{P}), t \in \mathcal{T} \quad (8)$$

where $q_{n,m,t}$ denotes the nominal water flow through pipeline n, m at time t and $\check{q}_{n,m,t}$ is an auxiliary variable which models the change in

water flow due to the affine response by water pumps as a linear function of uncertainty. Given the above description of water flows, the water balance constraint (2c) can be divided into its nominal and uncertainty-dependent terms which hold for any realization of uncertainty

$$\sum_{i \in N_{in}^J} q_{i,j,t} - \sum_{k \in N_{out}^J} q_{j,k,t} = d_{j,t} \quad j \in \mathcal{J}, t \in \mathcal{T} \quad (9a)$$

$$\sum_{i \in N_{in}^J} \check{q}_{i,j,t} - \sum_{k \in N_{out}^J} \check{q}_{j,k,t} = 0 \quad j \in \mathcal{J}, t \in \mathcal{T} \quad (9b)$$

Constraint (9a) ensures that the difference between incoming and outgoing water flows must be equal to the demand $d_{j,t}$ at any junction j and time t , while (9b) implies that the difference between incoming and outgoing adjustments in water flows in response to uncertainty must be equal to zero, given that the water demand $d_{j,t}$ is assumed to be fixed and therefore independent of uncertainty. Thus, constraints (9a) and (9b) ensure nominal and uncertainty-dependent water supply-demand match at all times. Similar to (8), pressure heads at the different junctions are modelled as

$$\tilde{h}_{j,t} = h_{j,t} + (\phi^T \Delta \omega_t) \check{h}_{j,t} \quad j \in \mathcal{J}, t \in \mathcal{T} \quad (10)$$

where $h_{j,t}$ denotes the nominal pressure head at junction j at time t and $\check{h}_{j,t}$ is an auxiliary variable which models the change in pressure head due to the response by water pumps as a linear function of uncertainty. Further, uncertainty propagates to the change in the water tank head as follows. The water storage tank head is described by constraints (2g) and (2h). By dividing (2g) and (2h) into their nominal and uncertainty-dependent counterparts, the set of constraints (11)–(11) is obtained which must hold for any realization of uncertainty

$$h_{s,t} - h_s^0 = \frac{\tau}{A_s} \left(\sum_{i \in N_{in}^s} q_{i,s,t} - \sum_{j \in N_{out}^s} q_{s,j,t} \right) \quad s \in \mathcal{S}, t = 1 \quad (11a)$$

$$h_{s,t} - h_{s,t-1} = \frac{\tau}{A_s} \left(\sum_{i \in N_{in}^s} q_{i,s,t} - \sum_{j \in N_{out}^s} q_{s,j,t} \right) \quad s \in \mathcal{S}, t > 1 \quad (11b)$$

$$\check{h}_{s,t} = \frac{\tau}{A_s} \left(\sum_{i \in N_{in}^s} \check{q}_{i,s,t} - \sum_{j \in N_{out}^s} \check{q}_{s,j,t} \right) \quad s \in \mathcal{S}, t = 1 \quad (11c)$$

$$\check{h}_{s,t} - \check{h}_{s,t-1} = \frac{\tau}{A_s} \left(\sum_{i \in N_{in}^s} \check{q}_{i,s,t} - \sum_{j \in N_{out}^s} \check{q}_{s,j,t} \right) \quad s \in \mathcal{S}, t > 1 \quad (11d)$$

where (11a)–(11d) describe the evolution of tank heads over time. As the initial tank head does not depend on uncertainty but rather on its initial level h_s^0 , at $t = 1$, the nominal part of constraint (2g) decomposes as (11a) while the uncertainty-dependent part decomposes as (11c). Differently, constraint (2h) is separated into its nominal and uncertainty-dependent parts (11b) and (11d) for $t > 1$. One can follow a similar approach to expand constraints (2d) and (2e). With the above description of water flows and nodal pressure heads, it is possible to expand the Darcy–Weisbach Eq. (2m) which describes friction head losses through water pipelines $(i, j) \in \mathcal{E}$ that do not contain a pump or a valve. Similarly, one can expand the pump hydraulic constraint (2o) describing the pump head pressure gain, as well as the pump energy consumption (2q). However, these constraints are nonlinear and nonconvex, which makes the problem computationally intractable. The convex relaxation of constraints (2m), (2o), and (2q) is explained in Section 3.

2.5. Availability cost

The proposed coordination mechanism is a centrally managed power and water market clearing problem that functions as a distribution-robust control mechanism for the management of wind power generation uncertainties. As outlined above, the DA market is cleared considering the predicted wind power forecasts as well as possible power imbalances that could result from the worst-case uncertainty realization. In response to uncertainty, dispatchable resources such as

flexible generators and pump loads can offer flexibility services to help compensate for power imbalances that could be realized at a later stage (i.e., RT stage). In this way, the DA decision is affected by the minimum and maximum capacity of the flexible assets and the flexibility they offer in RT. The objective is to achieve the highest social welfare in DA while leaving enough operational flexibility available in RT. Therefore, the proposed coordination mechanism is modelled as an augmented DA market clearing problem, in which the cost of flexibility utilization in RT is externalized from the DA social welfare. To regulate this mechanism, availability costs are defined for flexible generators C_e^{Av} and pump loads C_u^{Av} , which is the cost incurred by flexible assets to ensure the provision of flexibility services in RT.

The flexibility costs are defined as an affine function of the power and water control policy parameters. The total availability cost for generator e at time t is therefore given by $C_e^{Av} \alpha_{e,t}$ where $\alpha_{e,t} = (\phi^T \Delta \omega_t) \tilde{p}_{e,t}$ is an auxiliary variable used to model the amount of flexibility procured by generators. Similarly, the total availability cost for pump u at time t is $C_u^{Av} \beta_{u,t}$ where $\beta_{u,t} = (\phi^T \Delta \omega_t) \tilde{p}_{u,t}$ is an auxiliary variable used to model the amount of flexibility offered by water pumps.

3. Approximations and relaxations

3.1. Convex relaxation of Darcy–Weisbach equation

Since nonconvex problems are NP-hard and difficult to solve with available solvers, convex approximations and relaxations are implemented to make the problem computationally tractable. As mentioned above, the friction head losses in water pipes without pumps or valves are usually modelled with the Darcy–Weisbach Eq. (2m) which is nonconvex. The nonconvexity arises from the equality constraint that models head losses due to friction as a quadratic function of the flow rate. Ref. [24] overcomes this non-linearity by proposing a SOC relaxation, and by providing sufficient conditions for exactness. However, this relaxation approach does not apply to the proposed methodology due to the necessity of expanding constraint (2m) to describe uncertainty propagation. In particular, modelling friction head losses as a probabilistic chance constraint requires (2m) to be expressed as an inequality. Following the approach presented in [28] constraint (2m) is modelled as the intersection of a convex function and a concave function as shown in Fig. 2. Therefore, the feasible region enclosed between the resulting convex set can be modelled by inequalities (12a) and (12b) as follows

$$\tilde{h}_{i,t} - \tilde{h}_{j,t} \geq F_{i,j} q_{i,j,t}^2 \quad (i, j) \in \mathcal{E}, t \in \mathcal{T} \quad (12a)$$

$$\tilde{h}_{i,t} - \tilde{h}_{j,t} \leq r_{i,j}^0 + r_{i,j}^1 \tilde{q}_{i,j,t} \quad (i, j) \in \mathcal{E}, t \in \mathcal{T} \quad (12b)$$

where parameter $F_{i,j}$ represents the flow-independent part of the Darcy–Weisbach equation, while parameters $r_{i,j}^0$ and $r_{i,j}^1$ denote the y-intercept and the slope of the linear cut respectively. These parameters depend on the specific characteristic of pipe (i, j) such as the material, length and diameter. By first expanding and then separating uncertainty-dependent and independent counterparts of constraints (12a) and (12b), the following set of constraints (13) is obtained

$$\tilde{h}_{i,t} - \tilde{h}_{j,t} \geq F_{i,j} q_{i,j,t}^2 \quad (i, j) \in \mathcal{E}, t \in \mathcal{T} \quad (13a)$$

$$\tilde{h}_{i,t} - \tilde{h}_{j,t} \leq r_{i,j}^0 + r_{i,j}^1 \tilde{q}_{i,j,t} \quad (i, j) \in \mathcal{E}, t \in \mathcal{T} \quad (13b)$$

$$\phi^T \Delta \omega_t (\tilde{h}_{i,t} - \tilde{h}_{j,t}) \geq (\phi^T \Delta \omega_t) F_{i,j} 2 q_{i,j,t} \tilde{q}_{i,j,t} + (\phi^T \Delta \omega_t)^2 F_{i,j} \tilde{q}_{i,j,t}^2 \quad (i, j) \in \mathcal{E}, t \in \mathcal{T} \quad (13c)$$

$$\tilde{h}_{i,t} - \tilde{h}_{j,t} \leq r_{i,j}^1 \tilde{q}_{i,j,t} \quad (i, j) \in \mathcal{E}, t \in \mathcal{T} \quad (13d)$$

The convex hull relaxation approach can be used to convexify the nominal part of (12a) and (12b) into (13a) and (13b). However, this approach on its own is not sufficient to convexify the uncertainty-dependent counterpart of (12a). Expanding (12a) introduces new bilinear and quadratic terms that are dependent on uncertainty as can be seen in (13c).

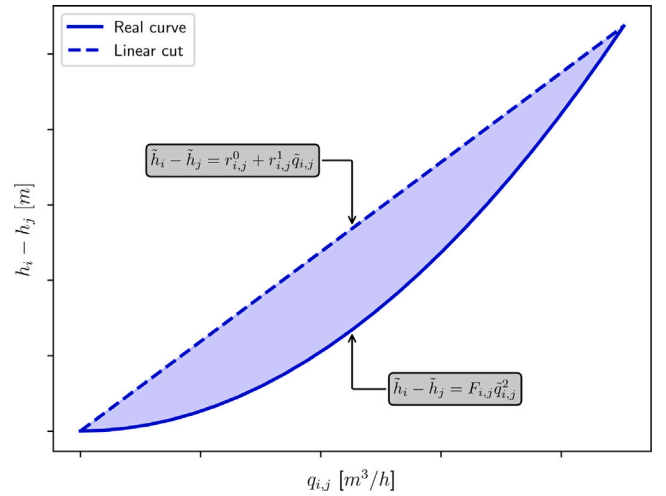


Fig. 2. Convex approximation of Darcy–Weisbach equation for pipe friction head losses; the continuous line is the real curve, the dashed line is the linear cut and the blue area is the resulting feasible region. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

To further convexify (13c), a novel hybrid approach is proposed consisting of a McCormick relaxation of the bilinear term $q_{i,j,t} \tilde{q}_{i,j,t}$ and a SOC relaxation of the quadratic term $\tilde{q}_{i,j,t}^2$. McCormick relaxation [43] consists of replacing the bilinear term with an auxiliary variable $m_{i,j,t}$ and the set of linear envelopes (14) for $(i, j) \in \mathcal{E}, t \in \mathcal{T}$ as follows

$$m_{i,j,t} \geq \max\{(q_{i,j,t}^L \tilde{q}_{i,j,t} + q_{i,j,t} \tilde{q}_{i,j,t}^L - q_{i,j,t}^L \tilde{q}_{i,j,t}^L), (q_{i,j,t}^U \tilde{q}_{i,j,t} + q_{i,j,t} \tilde{q}_{i,j,t}^U - q_{i,j,t}^U \tilde{q}_{i,j,t}^U)\} \quad (14a)$$

$$m_{i,j,t} \leq \min\{(q_{i,j,t}^U \tilde{q}_{i,j,t} + q_{i,j,t} \tilde{q}_{i,j,t}^L - q_{i,j,t}^U \tilde{q}_{i,j,t}^L), (q_{i,j,t}^L \tilde{q}_{i,j,t} + q_{i,j,t} \tilde{q}_{i,j,t}^U - q_{i,j,t}^L \tilde{q}_{i,j,t}^U)\} \quad (14b)$$

where $q_{i,j,t}^L$, $\tilde{q}_{i,j,t}^L$, $q_{i,j,t}^U$ and $\tilde{q}_{i,j,t}^U$ are parameters representing the estimated lower and upper bounds of variables $q_{i,j,t}$ and $\tilde{q}_{i,j,t}$ respectively. Lower and upper bounds can be obtained by minimizing and maximizing the variables of interest using a deterministic version of the program (1)–(3). Further, the quadratic term is relaxed by replacing $\tilde{q}_{i,j,t}^2$ with the auxiliary variable $\gamma_{i,j,t}$ and introducing one additional SOC constraint. This procedure leads to a tractable reformulation of (13c) as follows

$$\phi^T \Delta \omega_t (\tilde{h}_{i,t} - \tilde{h}_{j,t}) \geq (\phi^T \Delta \omega_t) F_{i,j} 2 m_{i,j,t} + (\phi^T \Delta \omega_t)^2 F_{i,j} \gamma_{i,j,t} \quad (i, j) \in \mathcal{E}, t \in \mathcal{T} \quad (15a)$$

$$\tilde{q}_{i,j,t}^2 \leq \gamma_{i,j,t} \quad (i, j) \in \mathcal{E}, t \in \mathcal{T} \quad (15b)$$

Therefore the nominal and uncertainty-dependent parts of the Darcy–Weisbach Eq. (2m) can be approximated with the set of constraints (13a), (13b), (13d), (14a), (14b), (15a), and (15b) that must hold for any realization of uncertainty. It is worth mentioning that, when dividing both sides of (15a) by $\phi^T \Delta \omega_t$, (15a) simplifies to a linear function of uncertainty (i.e., wind power deviation $\phi^T \Delta \omega_t$). It must be noted that the friction head losses calculated using the hybrid convex hull-based approach are not exact. Ref. [29] showed that it is possible to recover the exact values for the hydraulic heads under specific assumptions and network topology. The impact of the proposed convex hull relaxation is further discussed in Section 4.

3.2. Approximation of pump hydraulics and energy consumption

The hydraulic behaviour of a variable speed pump is usually modelled with a quadratic function of the flow rate (2o), which describes the pressure head gain imposed by pump i, j at time t . Constraint (2o) is nonlinear and nonconvex due to the equality sign as well as the

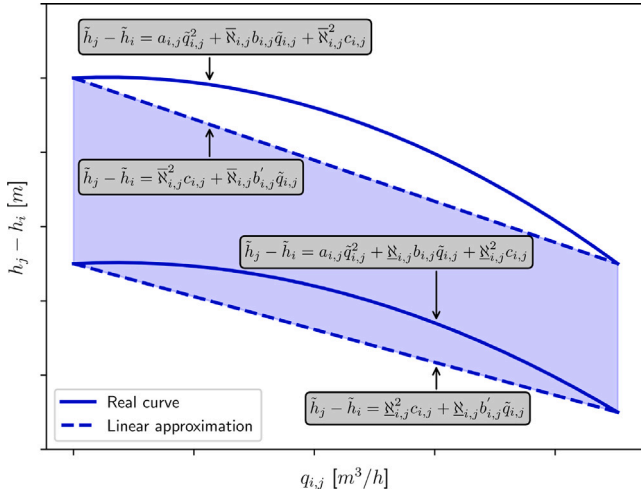


Fig. 3. Convex approximation of hydraulic behaviour of a variable speed pump; the continuous line is the real pump hydraulic curve, the dashed line is the linear approximation, and the blue area is the resulting feasible region. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

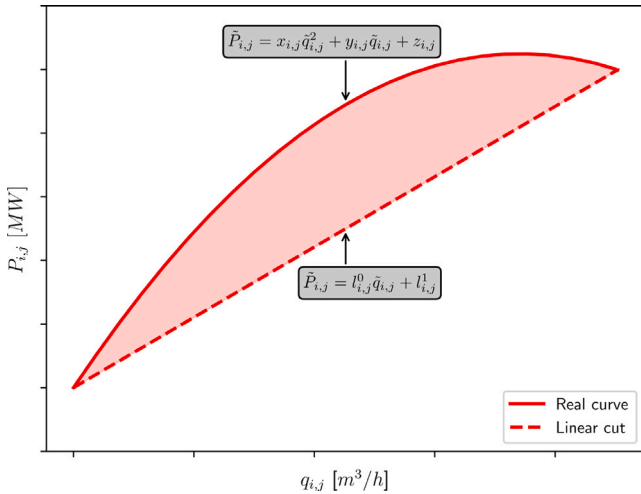


Fig. 4. Convex approximation of pump energy consumption of a variable speed pump; the continuous line is the real pump power curve, the dashed line is the linear cut, and the area in red is the resulting feasible region. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

negative coefficient $a_{i,j}$ in front of the quadratic term, which makes the quadratic function concave. Further, it must be noted that the hydraulic behaviour of a variable speed pump depends on the pump operating speed [44]. As shown in (20), the pump hydraulic function is corrected by a factor $\underline{\kappa}_{i,j} \leq \kappa_{i,j} \leq \bar{\kappa}_{i,j}$ expressing the ratio of the actual to nominal speed of pump i, j . $\underline{\kappa}_{i,j}$ and $\bar{\kappa}_{i,j}$ respectively represent the ratio of the lower- and the upper-bound to the nominal speed of the pump.

As is shown in Fig. 3, the feasible pump operating region is enclosed between the upper and lower pump hydraulic curves. Note that the upper and lower curves are a function of $\bar{\kappa}_{i,j}$ and $\underline{\kappa}_{i,j}$, respectively. Following a similar approach as in [28,29], the pump hydraulic function is approximated with a decreasing linear function as shown in Fig. 3. Thus, the feasible pump operating region is modelled with constraints (16a) and (16b). By expanding (16a) and (16b) into their nominal and

uncertainty-dependent counterparts, constraints (16c) and (16d) are obtained as follows

$$\tilde{h}_{j,t} - \tilde{h}_{i,t} \leq \bar{\kappa}_{i,j}^2 c_{i,j} - \bar{\kappa}_{i,j} b'_{i,j} \tilde{q}_{i,j,t} \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (16a)$$

$$\tilde{h}_{j,t} - \tilde{h}_{i,t} \geq \underline{\kappa}_{i,j}^2 c_{i,j} - \underline{\kappa}_{i,j} b'_{i,j} \tilde{q}_{i,j,t} \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (16b)$$

$$h_{j,t} - h_{i,t} + \phi^T \Delta \omega_t (\tilde{h}_{j,t} - \tilde{h}_{i,t}) \leq \bar{\kappa}_{i,j}^2 c_{i,j} - \bar{\kappa}_{i,j} b'_{i,j} (q_{i,j,t} + \phi^T \Delta \omega_t \tilde{q}_{i,j,t}) \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (16c)$$

$$h_{j,t} - h_{i,t} + \phi^T \Delta \omega_t (\tilde{h}_{j,t} - \tilde{h}_{i,t}) \geq \underline{\kappa}_{i,j}^2 c_{i,j} - \underline{\kappa}_{i,j} b'_{i,j} (q_{i,j,t} + \phi^T \Delta \omega_t \tilde{q}_{i,j,t}) \quad (i, j) \in \mathcal{P}, t \in \mathcal{T}. \quad (16d)$$

Parameters $b'_{i,j}$, $c_{i,j}$, $\bar{\kappa}_{i,j}$ and $\underline{\kappa}_{i,j}$ depend on the specific characteristics of the selected water pump (i, j) and can be calculated from pump characteristic curves and technical data sheets. Therefore, the nominal and uncertainty-dependent pump hydraulic response can be modelled with constraints (16c) and (16d). In what follows, the approximation of pump energy consumption is discussed.

The power required to run the electric motor of a variable-speed pump is usually modelled as a function of both pump flow and pressure head gain as shown in (2q). However, constraint (2q) contains nonlinear terms and is therefore nonconvex. Following a similar approach as in [28,29], constraint (2q) is replaced with a concave quadratic function of the flow rate (17a) as shown in Fig. 4. It must be noted that by expressing pump energy consumption as a quadratic function of the flow rate, we account for the flow and head relationship as well as the effect of varying electric motor efficiency on the power consumption of a variable speed pump. Following the convex hull relaxation approach as above, constraint (17a) is modelled as the intersection of a concave inequality (17b) and a convex inequality (17c) as follows

$$\tilde{P}_{i,j,t} = x_{i,j} \tilde{q}_{i,j,t}^2 + y_{i,j} \tilde{q}_{i,j,t} + z_{i,j} \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (17a)$$

$$\tilde{P}_{i,j,t} \leq x_{i,j} \tilde{q}_{i,j,t}^2 + y_{i,j} \tilde{q}_{i,j,t} + z_{i,j} \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (17b)$$

$$\tilde{P}_{i,j,t} \geq l_{i,j}^0 \tilde{q}_{i,j,t} + l_{i,j}^1 \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (17c)$$

where $x_{i,j}$, $y_{i,j}$, $z_{i,j}$, $l_{i,j}^0$, and $l_{i,j}^1$ are empirical parameters that depend on the specific characteristics of water pump (i, j) and can be calculated from pump power curves obtained from technical data sheets. Separating uncertainty-dependent and independent terms in (17b) and (17c) the set of constraints (18) is obtained as follows.

$$P_{i,j,t} \leq x_{i,j} q_{i,j,t}^2 + y_{i,j} q_{i,j,t} + z_{i,j} \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (18a)$$

$$P_{i,j,t} \geq l_{i,j}^0 q_{i,j,t} + l_{i,j}^1 \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (18b)$$

$$(\phi^T \Delta \omega_t) \tilde{P}_{i,j,t} \leq (\phi^T \Delta \omega_t) 2x_{i,j} \tilde{q}_{i,j,t} q_{i,j,t} + (\phi^T \Delta \omega_t)^2 x_{i,j} \tilde{q}_{i,j,t}^2 + (\phi^T \Delta \omega_t) y_{i,j} \tilde{q}_{i,j,t} \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (18c)$$

$$\tilde{P}_{i,j,t} \geq l_{i,j}^0 \tilde{q}_{i,j,t} \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (18d)$$

As can be seen, constraint (18c) contains bilinear and quadratic terms that are dependent on uncertainty. A similar hybrid approach as introduced in Section 3.1 is adopted to further convexify (18c). Firstly, McCormick relaxation is used to convexify the bilinear term $q_{i,j,t} \tilde{q}_{i,j,t}$. Secondly, SOC relaxation is applied to approximate the quadratic term $\tilde{q}_{i,j,t}^2$. The McCormick envelopes are given by the set of constraints (19) for $(i, j) \in \mathcal{P}, t \in \mathcal{T}$ as follows

$$m_{i,j,t} \geq \max\{(q_{i,j,t}^L \tilde{q}_{i,j,t} + q_{i,j,t} \tilde{q}_{i,j,t}^L - q_{i,j,t}^L \tilde{q}_{i,j,t}^L), (q_{i,j,t}^U \tilde{q}_{i,j,t} + q_{i,j,t} \tilde{q}_{i,j,t}^U - q_{i,j,t}^U \tilde{q}_{i,j,t}^U)\} \quad (19a)$$

$$m_{i,j,t} \leq \min\{(q_{i,j,t}^U \tilde{q}_{i,j,t} + q_{i,j,t} \tilde{q}_{i,j,t}^U - q_{i,j,t}^U \tilde{q}_{i,j,t}^U), (q_{i,j,t}^L \tilde{q}_{i,j,t} + q_{i,j,t} \tilde{q}_{i,j,t}^L - q_{i,j,t}^L \tilde{q}_{i,j,t}^L)\} \quad (19b)$$

where $q_{i,j,t}^L$, $\tilde{q}_{i,j,t}^L$, $q_{i,j,t}^U$ and $\tilde{q}_{i,j,t}^U$ are parameters representing the estimated upper and lower bounds of variables $q_{i,j,t}$ and $\tilde{q}_{i,j,t}$ respectively. Further, the SOC relaxation consists of replacing $\tilde{q}_{i,j,t}^2$ with the auxiliary variable $\zeta_{i,j,t}$ and introducing one additional (SOC) constraint, which leads to a tractable reformulation of (18c) as follows

$$\begin{aligned} (\phi^T \Delta \omega_t) \tilde{p}_{i,j,t} &\leq \\ (\phi^T \Delta \omega_t) 2x_{i,j} m_{i,j,t} + (\phi^T \Delta \omega_t)^2 x_{i,j} \zeta_{i,j,t} + (\phi^T \Delta \omega_t) y_{i,j} \tilde{q}_{i,j,t} \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \end{aligned} \quad (20a)$$

$$\tilde{q}_{i,j,t}^2 \leq \zeta_{i,j,t} \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (20b)$$

where $m_{i,j,t}$ is the McCormick auxiliary variable used to replace $q_{i,j,t} \tilde{q}_{i,j,t}$, and $\zeta_{i,j,t}$ is the auxiliary variable used to replace $\tilde{q}_{i,j,t}^2$ in (18c). Therefore the pump energy consumption expressed in (2q) can be approximated with the set of convex constraints (18a), (18b), (18d), (19a), (19b), (20a), and (20b). Here it is vital to note that the summation of nominal and uncertainty-dependent pump energy consumption terms $P_{i,j,t} + (\phi^T \Delta \omega_t) \tilde{p}_{i,j,t}$ must be bounded between the minimum and maximum value of $x_{i,j} q_{i,j,t}^2 + y_{i,j} q_{i,j,t} + z_{i,j}$ for the given operating flow range of pump i, j . Further, it must be noted that, as the objective function maximizes utility and the pump load bids are fixed to the maximum auction price (i.e., price cap), the convex hull relaxation of the pump energy consumption is exact. So far, affine control policies have been defined, and uncertainty propagation has been described by first expanding the constraints, and then separating their nominal and uncertainty-dependent counterparts. Additionally, tractable reformulations of all nonconvex constraints have been proposed. Section 3.3 provides the convex reformulation of the stochastic program (1)–(3).

3.3. Power and water coordination with affine policies

In the following, the complete reformulation of the DRCC program (1)–(3) is provided, based on the assumption of affine response to uncertainty and the convex relaxations and approximations described in the previous sections. The reformulation of the DRCC problem takes on the following form

$$\begin{aligned} \min_{\Xi, \mathbb{P} \in \Pi} & \left[\left(\sum_{e \in \mathcal{G}} C_e^G (p_{e,t}^G) - \sum_{l \in \mathcal{D}} U_l^D p_{l,t}^D - \sum_{u \in \mathcal{P}} U_u^P p_{u,t}^P \right) \right. \\ & \left. + \sum_{e \in \mathcal{G}} C_e^{Av} \alpha_{e,t} + \sum_{u \in \mathcal{P}} C_u^{Av} \beta_{u,t} \right] \end{aligned} \quad (21a)$$

s.t.

Power constraints

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[p_{e,t}^G + (\phi^T \Delta \omega_t) \tilde{p}_{e,t}^G \leq \bar{p}_e^G] \geq (1 - \epsilon) \quad e \in \mathcal{G}, t \in \mathcal{T} \quad (21b)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[p_{e,t}^G + (\phi^T \Delta \omega_t) \tilde{p}_{e,t}^G \geq \underline{p}_e^G] \geq (1 - \epsilon) \quad e \in \mathcal{G}, t \in \mathcal{T} \quad (21c)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[f_{n,m,t} + \phi^T \Delta \omega_t [E_0^2 B_{n,m} (\tilde{\theta}_{n,t} - \tilde{\theta}_{m,t})] \leq \bar{f}_{n,m}] \geq (1 - \epsilon) \quad (n, m) \in \mathcal{L}, t \in \mathcal{T} \quad (21d)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[f_{n,m,t} + \phi^T \Delta \omega_t [E_0^2 B_{n,m} (\tilde{\theta}_{n,t} - \tilde{\theta}_{m,t})] \geq \underline{f}_{n,m}] \geq (1 - \epsilon) \quad (n, m) \in \mathcal{L}, t \in \mathcal{T} \quad (21e)$$

$$\alpha_{e,t} = (\phi^T \Delta \omega_t) \tilde{p}_{e,t} \quad e \in \mathcal{G}, t \in \mathcal{T} \quad (21f)$$

$$(1b), (6), (7) \quad (21g)$$

Water constraints

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[h_{j,t} + (\phi^T \Delta \omega_t) \tilde{h}_{j,t} \geq \underline{H}_j + H_j^0] \geq (1 - \epsilon) \quad j \in \mathcal{J}, t \in \mathcal{T} \quad (22a)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[h_{j,t} + (\phi^T \Delta \omega_t) \tilde{h}_{j,t} \leq \bar{H}_j + H_j^0] \geq (1 - \epsilon) \quad j \in \mathcal{J}, t \in \mathcal{T} \quad (22b)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[h_{s,t} + (\phi^T \Delta \omega_t) \tilde{h}_{s,t} \geq \underline{H}_s + H_s^0] \geq (1 - \epsilon) \quad s \in \mathcal{S}, t \in \mathcal{T} \quad (22c)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[h_{s,t} + (\phi^T \Delta \omega_t) \tilde{h}_{s,t} \leq \bar{H}_s + H_s^0] \geq (1 - \epsilon) \quad s \in \mathcal{S}, t \in \mathcal{T} \quad (22d)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[\tilde{h}_{s,t} + (\phi^T \Delta \omega_t) \tilde{h}_{s,t} \geq \underline{H}_s^{end}] \geq (1 - \epsilon) \quad s \in \mathcal{S}, t = end \quad (22e)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[\tilde{h}_{s,t} + (\phi^T \Delta \omega_t) \tilde{h}_{s,t} \leq \bar{H}_s^{end}] \geq (1 - \epsilon) \quad s \in \mathcal{S}, t = end \quad (22f)$$

$$h_{j,t} - h_{i,t} = K_{i,j,t} \quad (i, j) \in \mathcal{V}, t \in \mathcal{T} \quad (22g)$$

$$\tilde{h}_{j,t} - \tilde{h}_{i,t} = \tilde{K}_{i,j,t} \quad (i, j) \in \mathcal{V}, t \in \mathcal{T} \quad (22h)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[q_{i,j,t} + (\phi^T \Delta \omega_t) \tilde{q}_{i,j,t} \leq \bar{Q}_{i,j}] \geq (1 - \epsilon) \quad (i, j) \in (\mathcal{E} \cup \mathcal{V} \cup \mathcal{P}), t \in \mathcal{T} \quad (22i)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[q_{i,j,t} + (\phi^T \Delta \omega_t) \tilde{q}_{i,j,t} \geq \underline{Q}_{i,j}] \geq (1 - \epsilon) \quad (i, j) \in (\mathcal{E} \cup \mathcal{V} \cup \mathcal{P}), t \in \mathcal{T} \quad (22j)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[\tilde{h}_{j,t} - \tilde{h}_{i,t} \geq F_{i,j} 2m_{i,j,t} + (\phi^T \Delta \omega_t) F_{i,j} \gamma_{i,j,t}] \geq (1 - \epsilon) \quad (i, j) \in \mathcal{E}, t \in \mathcal{T} \quad (22k)$$

$$\begin{aligned} \min_{\mathbb{P} \in \Pi} \mathbb{P}[h_{j,t} - h_{i,t} + \phi^T \Delta \omega_t (\tilde{h}_{j,t} - \tilde{h}_{i,t}) \leq \\ \tilde{\kappa}_{i,j}^2 c_{i,j} - \tilde{\kappa}_{i,j} b'_{i,j} (q_{i,j,t} + \phi^T \Delta \omega_t \tilde{q}_{i,j,t})] \geq (1 - \epsilon) \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \end{aligned} \quad (22l)$$

$$\begin{aligned} \min_{\mathbb{P} \in \Pi} \mathbb{P}[h_{j,t} - h_{i,t} + \phi^T \Delta \omega_t (\tilde{h}_{j,t} - \tilde{h}_{i,t}) \geq \\ \kappa_{i,j}^2 c_{i,j} - \kappa_{i,j} b'_{i,j} (q_{i,j,t} + \phi^T \Delta \omega_t \tilde{q}_{i,j,t})] \geq (1 - \epsilon) \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \end{aligned} \quad (22m)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[\tilde{p}_{i,j,t} \leq 2x_{i,j} m_{i,j,t} + (\phi^T \Delta \omega_t) x_{i,j} \zeta_{i,j,t} + y_{i,j} \tilde{q}_{i,j,t}] \geq (1 - \epsilon) \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (22n)$$

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[\underline{P}_{i,j} \leq P_{i,j,t} + (\phi^T \Delta \omega_t) \tilde{p}_{i,j,t} \leq \bar{P}_{i,j}] \geq (1 - \epsilon) \quad (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (22o)$$

$$\beta_{u,t} = (\phi^T \Delta \omega_t) \tilde{p}_{u,t} \quad p \in \mathcal{P}, t \in \mathcal{T} \quad (22p)$$

$$(2b), (9), (11), (13a), (13b), (13d), (14), (15b), (18a), (18b), (18d), (19), (20b) \quad (22q)$$

Coupling constraints

$$p_{u,t}^P = P_{i,j,t}, \quad u \in \Psi_{P_{i,j}}, (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (23a)$$

$$\tilde{p}_{u,t}^P = \tilde{p}_{i,j,t}, \quad u \in \Psi_{P_{i,j}}, (i, j) \in \mathcal{P}, t \in \mathcal{T} \quad (23b)$$

Where $\Xi = p_{i,t}^D, p_{e,t}^G, \tilde{p}_{e,t}^G, p_{u,t}^P, \tilde{p}_{u,t}^P, \theta_{n,t}, \tilde{\theta}_{n,t}, f_{n,m,t}, \tilde{f}_{n,m,t}, h_{i,t}, \tilde{h}_{i,t}, h_{r,t}, q_{i,j,t}, \tilde{q}_{i,j,t}, h_{s,t}, \tilde{h}_{s,t}, P_{i,j,t}, \tilde{P}_{i,j,t}, K_{i,j,t}, \tilde{K}_{i,j,t}, m_{i,j,t}, \gamma_{i,j,t}, \zeta_{i,j,t}, \alpha_{e,t}, \beta_{u,t}$ is the set of nominal and uncertainty-dependent optimization decision variables. Introducing constraints (22o) ensures that the change in energy consumption due to pump response stays within the pump's operational limits. Parameters $\underline{P}_{i,j}$ and $\bar{P}_{i,j}$ represent the lower and upper bounds for pump energy consumption. Further, it must be noted that under the assumption of zero mean forecast error, the expectation term in the objective function (1a) can be removed. Thus, objective (1a) simplifies to (21a). For further details, the reader is referred to [10,31]. Nevertheless, at this stage, the problem is still intractable. As explained in [31], the requirement to model the constraints as probabilistic DRCCs makes the problem (21)–(23) infinite-dimensional. A finite and tractable reformulation of DRCCs is proposed in Section 3.4.

3.4. SOC reformulation of probabilistic constraints

Depending on the system under study, the vector of random variables $\Delta \omega \in \mathbb{R}^Z$ might follow different probability distributions. When only limited information is known about the distribution of stochastic random variables it is appropriate to consider a set of probability distributions (a.k.a, ambiguity set), as previously defined in Section 2.1, and only assume certain properties of the distribution. Different assumptions will lead to different analytical reformulations of the DRCCs as explained in [30]. These reformulations are valid for all distributions that share the same common properties. Under the assumption of known (and finite) first and second-order moments (i.e., mean and covariance), individual chance constraints can be approximated as SOC constraints using Chebyshev's Inequality as explained in [30]. As an example, let us consider the DRCC (21c):

$$\min_{\mathbb{P} \in \Pi} \mathbb{P}[p_{e,t}^G + (\phi^T \Delta \omega_t) \tilde{p}_{e,t}^G \geq \underline{p}_e^G] \geq (1 - \epsilon) \quad e \in \mathcal{G}, t \in \mathcal{T} \quad (24)$$

where $\Delta \omega_t$ is a vector of random variables and ϕ^T is a vector of ones. Here, $p_{e,t}^G$ is the nominal generation output and $\tilde{p}_{e,t}^G$ is a zero mean fluctuating component modelled as an affine function of the wind

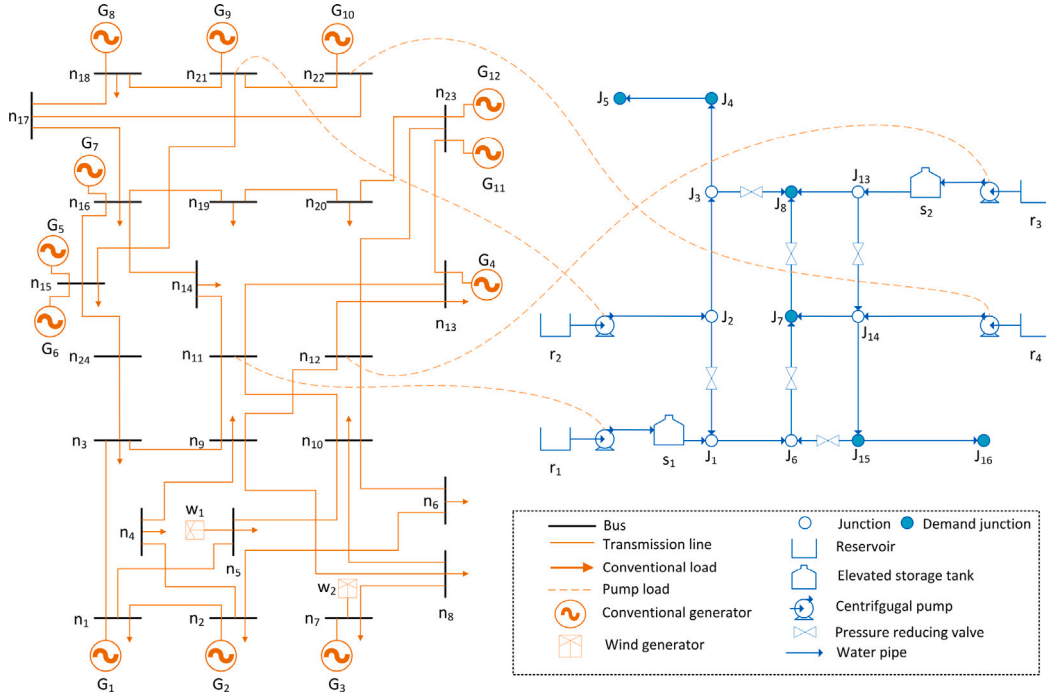


Fig. 5. Coupled power network (left) and water network (right). The orange dashed lines show where the water pumps are connected to the power network. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

forecast error. Thus, the second term in (24) expresses the influence of uncertainty on the respective constraint. Given the above description of uncertainty, the probabilistic DRCC can be approximated with the following deterministic SOC constraint (25a). The first term $(\tilde{p}_{e,t}^G)\mu_t$, which can be either positive or negative, is a correction factor related to the direction of the mean forecast error bias (i.e., surplus or deficit). The second term expresses the worst-case wind forecast deviation and can be interpreted as a reliability margin against uncertainty. This term is always positive, meaning that it always relates to a reduction in the available wind power capacity. Under the assumption of zero mean forecast error, constraint (25a) reduces to (25b).

$$-p_{e,t}^G + \underline{p}_e^G \leq (\tilde{p}_{e,t}^G)\mu_t + \sqrt{\frac{1-\epsilon}{\epsilon}} \|(\tilde{p}_{e,t}^G)\Sigma_t^{1/2}\|_2 \quad e \in \mathcal{G}, t \in \mathcal{T} \quad (25a)$$

$$-p_{e,t}^G + \underline{p}_e^G \leq \sqrt{\frac{1-\epsilon}{\epsilon}} \|(\tilde{p}_{e,t}^G)\Sigma_t^{1/2}\|_2 \quad e \in \mathcal{G}, t \in \mathcal{T} \quad (25b)$$

The other probabilistic DRCCs of the program (21)–(23) are reformulated following a similar approach as in (25). Notably, the reformulation of chance constraints as Second-Order Cone (SOC) constraints allows us to transform the infinite-dimensional probabilistic problem into an equivalent deterministic one. This transformation is achieved by representing the random vector $\Delta\omega_t$ with the covariance matrix, which is a fixed input to the final problem. Ref. [45] explains that, when using the SOC reformulation based on Chebyshev's Inequality, parameter ϵ should be carefully selected. In particular, by selecting smaller values for ϵ (i.e., $\epsilon = 0.01$) the problem becomes increasingly robust. In this study, an ad hoc trial-and-error process is used to ensure that sufficiently large values of ϵ are chosen to avoid ultra-conservative solutions. Without loss of generality, individual end-users have the flexibility to adjust the parameter ϵ , tailoring the individual chance constraints to meet the specific requirements of the given system.

3.5. SOC reformulation of flexibility functions

With the above description of uncertainty, constraint (21f), describing the amount of flexibility provided by conventional power generators, and constraint (22p), describing the flexibility provided by water pumps, can be reformulated as a function of the norm of the power and water control policy parameters. Thus, constraints (21f) and (22p) can be approximated with the set of constraints (26), which leads to a tractable reformulation of the program (21)–(23).

$$\alpha_{e,t} \geq \|(\tilde{p}_{e,t}^G)\Sigma_t^{1/2}\|_2 \quad e \in \mathcal{G}, t \in \mathcal{T} \quad (26a)$$

$$\beta_{u,t} \geq \|(\tilde{p}_{u,t}^P)\Sigma_t^{1/2}\|_2 \quad u \in \mathcal{P}, t \in \mathcal{T} \quad (26b)$$

Here $\alpha_{e,t}$ is an auxiliary variable used to model the amount of flexibility offered by generator e at time t . Similarly, $\beta_{u,t}$ is an auxiliary variable used to model the amount of flexibility offered by water pump u at time t . Since the objective function minimizes the total availability cost incurred by the system, the SOC relaxation of the availability functions (26a) and (26b) is exact. In Section 4, the applicability of the proposed model is demonstrated through a case study, and the impact of the proposed relaxations and approximations is discussed.

4. Results and discussion

In this section, the applicability of the proposed methodology is demonstrated with a case study in which the central market operator solves the augmented DA market clearing problem for the coupled power–water network. The market operator is assumed to have full access to all the necessary information for the control of the integrated system. Firstly, the case study setup is described. Secondly, the numerical results of the model are presented. Thirdly, sensitivity analysis is performed to assess the impact of different availability costs on the model results. Lastly, the impact of the proposed relaxations and approximations on the accuracy of the results is analysed.

4.1. Case study

The case study is based on the coupled power and water network shown in Fig. 5. The Power Transmission Network (PTN) is based on a modified version of the IEEE 24-Bus Reliability Test System presented in [46]. The power network consists of 12 conventional generators, 2 wind farms with a maximum capacity of 50 MW each, 17 conventional loads, and 4 pump loads with a maximum power output of 12 MW each. The different buses are connected with a set of 34 power transmission lines. The slack bus is located at node 24. The water network is an extension of the one presented in [47]. It consists of 20 nodes including 12 junctions, 4 reservoirs, and 2 water storage tanks. Similar to [24,29], water storage tanks are modelled as two-node elements with the inlet located at the top of the tank and the outlet at the bottom. The different nodes are connected by 22 edges consisting of 10 water pipes, 6 pressure-reducing valves, and 4 pumps. The remaining two edges are imaginary connections that represent the virtual link between the inlet and outlet of the water storage tanks. A set of 2000 wind forecast scenarios is used to estimate the covariance matrix of wind forecast errors based on empirical data obtained from [48]. The violation probability ϵ is fixed at a value of 0.1 for all DRCCs. The availability cost, which is the cost incurred by the flexible assets to provide the flexibility service, is set to 500 €/MW for conventional generators (i.e., C_e^{Av}) and 5 €/MW for water pumps (i.e., C_u^{Av}). In the base case, the availability cost of pumps is significantly lower than the availability cost of generators. In Section 4.3, a local sensitivity analysis is performed to assess the impact of different availability costs on the optimal solution. Additional parameter information about the configuration of the test case is provided in Table A.1 to Table A.5 in Appendix. The problem is implemented in Julia v1.5.3, modelled with JuMP v0.21.6, and solved with Mosek v1.2.1. The problem is solved with an average CPU time of 2.67 s using a laptop with 32 GB RAM with a Processor Intel(R) Xeon(R) 6 Core(s) running at 2.80 GHz. The solution to the problem provides the nominal DA schedule for the integrated power and water network as well as the uncertainty-dependent decisions that characterize the response to uncertainty in wind power generation.

4.2. Numerical results

Fig. 6 shows the optimal DA dispatch schedule resulting from the coordinated operation of power and water networks for the baseline case, in which the selected violation probability is $\epsilon = 0.1$ and the availability costs are $C_e^{Av} = 500$ €/MW for generators and $C_u^{Av} = 5$ €/MW for pumps. The conventional power load (i.e., load that is not associated with water pumps) is shown in green, the water pump load is shown in blue, and the power dispatched by conventional generators (i.e., all generators excluding wind turbines) is indicated by the black dashed line. The largest share of the total load is met by conventional generators. The remaining portion is met by wind generators. While the conventional load largely varies throughout the 24 h, the pump load remains relatively stable. This aligns with the assumption of fixed water demands in the system. An hourly water demand profile would lead to a larger variation in water pump loads. Further, it is observed that pump energy consumption tends to be maximized, which has a limiting effect on the change in pump load throughout time. This is in line with the assumption that pump bids are set at the price cap. Fig. 7 presents the scheduled DA volumes for water storage tank 1, indicated by the green line, and tank 2, indicated by the blue line. Additionally, the DA wind power forecast and the actual (i.e., real-time) wind power realization are shown on the secondary axis by the red dashed line and the purple dashed line respectively. The results show that the volume in the storage tanks varies during the 24 h in response to changes in total power load as well as wind power generation. That is, water is pumped into elevated storage tanks at times when energy generation is cheaper. For instance, between hours 1 and 8 the volume of both

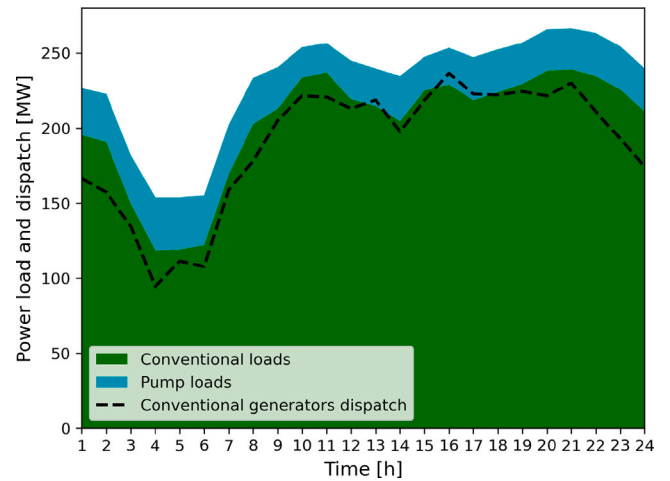


Fig. 6. Scheduled DA power dispatch highlighting conventional power loads (green area), pump loads (blue area), and power supplied by conventional generators (black dashed line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

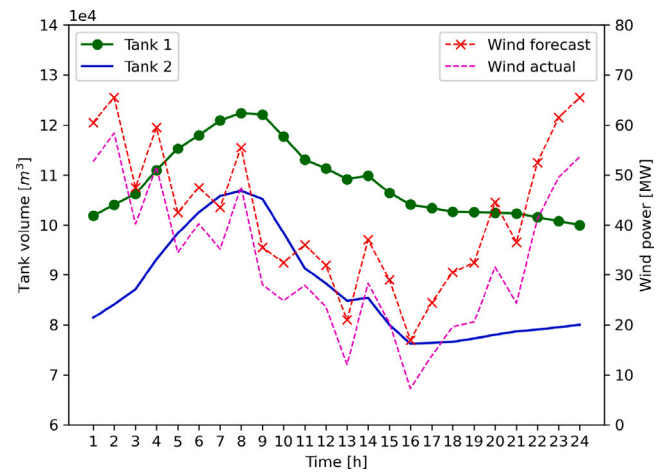


Fig. 7. Scheduled DA water volume in storage tank 1 (continuous green line with circle marker), storage tank 2 (continuous blue line), DA wind power forecast (dashed red line with cross marker), and RT wind power generation (dashed purple line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

storage tanks increases. This is because the load on the power network is relatively low and the availability of wind power that comes at zero marginal production cost is relatively high. Differently, between hours 8 and 16, the load on the power network is relatively high and wind power production is relatively low. In response, the water network uses elevated storage tanks to supply a significant amount of water through gravitational potential energy, i.e., between hour 8 and hour 16, which results in a decrease in water tank volume. This enables the system to reduce the electricity demand for pumping and, thus, the overall power load on the network. The flexible operation of water pumps and water storage tanks is beneficial because it prevents the dispatch of more expensive generating units, i.e., generators with a higher marginal production cost. These results show that the smart operation of the water system as a DR provider can effectively contribute to achieving higher economic efficiency.

Affine policies are assigned to flexible generators and water pumps on top of their nominal DA schedule to compensate for wind power

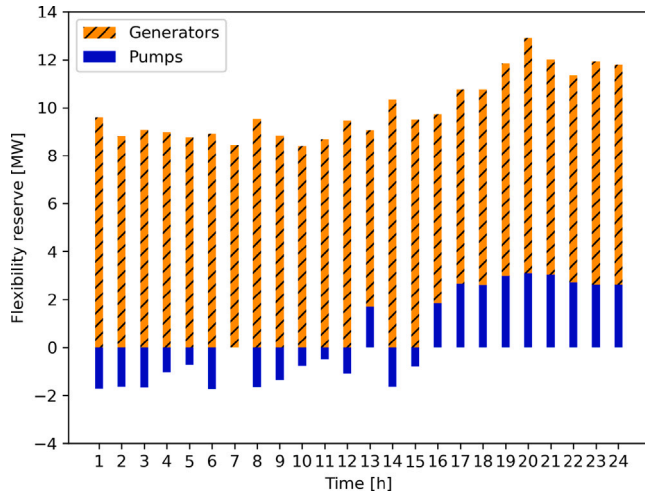


Fig. 8. Flexibility reserves assigned to conventional generators (orange) and water pumps (blue); a negative pump flexibility reserve means that pumps may increase their electricity consumption. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

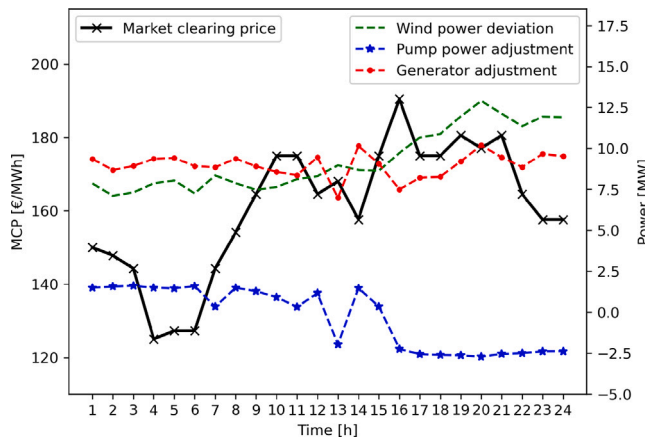


Fig. 9. Market-clearing prices (black solid line with cross marker), absolute wind power deviation (green dashed line), pump flexibility response (blue dashed line with star marker) and generator flexibility response (red dashed line with point marker). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

deviations that might occur during the RT stage. These policies are assigned such that enough flexibility is reserved to ensure a stable operation of the power system in RT. Fig. 8 shows the provision of flexibility reserves by flexible generators in orange and water pumps in blue. The bars indicate the amount of flexibility provided by the different assets. As a convention, a positive generation reserve means that generators may increase their production, while a positive pump reserve implies that water pumps may reduce their electricity consumption. Thus, a negative pump reserve implies that water pumps may increase their flow rate and thus their electricity consumption. The flexibility provided by generators and pumps should always sum up to the total wind power deviation. This means that the power mismatch is fully compensated. Results show that a major share of the total wind power deviation is compensated by flexible generators at all times. Water pumps can only compensate for power system imbalances up to a maximum of 25% of the total wind power mismatch as can be seen at hour 20. This is in line with previous assumptions

on the physical characteristics of the water system and the associated hydraulic constraints. Further, it is observed that, at certain times, generators increase their power output and, at the same time, water pumps increase their power consumption, i.e., at hours 1 to 6 as well as hours 8 to 12 and hours 14 to 15. Differently, at hours 13 as well as 17 to 24, generators increase their power output while pumps decrease their power consumption. This suggests that the response of flexible assets does not always go in the same direction with respect to the system's needs at a specific time. This could indicate a cross-temporal flexibility exchange taking place between flexible generators and water pumps. This issue is further investigated in Section 4.3.

Solving the DA market-clearing problem provides information about the market-clearing prices (MCPs) as well as the impact of the proposed affine policies on the overall system utility. It is known that the dual variable associated with the power balance constraint (6a) in the DA market gives the market clearing price. The dual variable represents the shadow price, or in other words the instantaneous change in the objective value that would be incurred by changing the right-hand side of constraint (6a) by one unit. Fig. 9 shows the resulting MCPs in black, the total wind power deviation in green, the aggregated flexibility executed by generators in red, and the aggregated flexibility executed by water pumps in blue. The difference between the red and the blue line gives the green line (i.e., wind power deviation). By comparing Fig. 9 with Fig. 6, it is observed that the MCP varies throughout the day depending on the availability of wind power as well as the overall load on the network. Between hours 1 and 8, the MCPs fluctuate between 125 and 155 €/MWh, which is relatively low compared to the MCPs observed between hours 9 and 21. The lower prices at the beginning of the day are due to the lower overall load on the network as well as the higher availability of wind power. When the load increases and wind power production reduces, the MCPs become relatively higher. Towards the end of the day, i.e. between hours 22 and 24, the MCPs decrease again as a result of the rising wind power production. Further, it is observed that pump power consumption response is inversely correlated with wind power deviation and MCPs. That is, water pumps increase their energy consumption in RT when wind power deviation and MCP are relatively low and vice versa. In particular, water pumps increase their power consumption at the beginning of the time horizon, i.e. between hours 1 and 12, while they reduce their power consumption towards the end of the day, i.e. between hours 15 and 24. Further, results show that between hours 1 and 12 the aggregated flexibility response of generators is larger than the total wind power deviation. At these hours, the additional flexibility provided by generators is utilized by the water pumps to increase their power consumption and thus their flow rate. Results suggest that, as wind power deviation and MCPs increase, the value of delivering pump flexibility (i.e., reducing pump load) becomes higher. However, this interpretation should be verified and supported with additional information. Further discussion is provided in Section 4.3.

4.3. Sensitivity analysis

Local sensitivity analysis is performed to assess the impact of different availability costs on the optimal solution. The sensitivity analysis consists of iteratively increasing and decreasing the availability cost of generators C_e^{Av} and pumps C_u^{Av} in incremental steps, to assess the sensitivity of the model to different choices of availability costs. Different scenarios are modelled in which parameters C_e^{Av} and C_u^{Av} are varied between a value of 5 €/MW and 1000 €/MW as shown in Table 2. As an example, scenario P50 G100 indicates that the availability cost is 50 €/MW for all pumps (P) and 100 €/MW for all generators (G). The results are compared in terms of the total revenue of generators in DA (R^{G-DA}), total revenue of generators in RT (R^{G-RT}), energy costs incurred by water pumps in DA (C^{P-DA}), and revenue of water pumps in RT (R^{P-RT}). All values in Table 2 are given in thousands of euros per day (K€/day). The results indicate that R^{G-DA} exhibits slight variations

Table 2

Model sensitivity to different availability costs for all scenarios of analysis. Values are expressed in thousands of euros per day (K€/day).

Scenario	^a $R^{G,DA}$	^b $R^{G,RT}$	^c $C^{P,DA}$	^d $R^{P,RT}$
P5 G5	742.81	1.10	108.19	0.11
P5 G50	742.74	10.91	108.38	0.12
P5 G100	742.88	21.75	108.62	0.14
P5 G500	742.86	106.77	108.25	0.21
P5 G1000	745.14	212.86	107.75	0.24
P50 G5	741.79	1.11	106.43	0.00
P50 G50	741.79	11.13	106.43	0.02
P50 G100	741.86	22.17	106.78	0.24
P50 G500	743.34	107.18	107.85	1.66
P50 G1000	743.34	213.50	108.43	1.96
P100 G5	741.79	1.11	106.43	0.00
P100 G50	741.79	11.14	106.43	0.00
P100 G100	741.79	22.28	106.43	0.00
P100 G500	742.42	108.30	106.49	2.12
P100 G1000	742.76	213.78	107.92	3.60
P500 G5	741.79	1.11	106.43	0.00
P500 G50	741.79	11.14	106.43	0.00
P500 G100	741.79	22.28	106.43	0.00
P500 G500	741.79	111.41	106.43	0.00
P500 G1000	739.94	220.97	104.99	0.93
P1000 G5	741.79	1.11	106.43	0.00
P1000 G50	741.79	11.14	106.43	0.00
P1000 G100	741.79	22.28	106.43	0.00
P1000 G500	741.79	111.41	106.43	0.00
P1000 G1000	741.79	222.81	106.43	0.00

$$^a R^{G,DA} = \sum (MCP_i p_{e,i}^G).$$

$$^b R^{G,RT} = \sum (C_e^{AV} \alpha_{e,i}).$$

$$^c C^{P,DA} = \sum (MCP_i p_{u,i}^P).$$

$$^d R^{P,RT} = \sum (C_u^{AV} \beta_{u,i}).$$

across different scenarios, ranging from a minimum of 739.94 K€/day in Scenario P500 G1000 to a maximum of 745.14 K€/day in Scenario P5 G1000. In contrast, the revenue of generators in RT ($R^{G,RT}$) is significantly influenced by the choice of availability costs. The lowest revenue in RT occurs in Scenario P5 G5 at 1.10 K€/day, while the highest revenue is achieved in Scenario P1000 G1000 at 222.81 K€/day. The energy-related cost of pumping operations in DA ($C^{P,DA}$) remains relatively stable, fluctuating between a minimum of 104.99 K€/day in Scenario P500 G1000 and a maximum of 108.62 K€/day in Scenario P5 G100. On the other hand, $R^{P,RT}$, indicating the revenue of water pumps in RT, exhibits substantial variations across different scenarios. However, it remains relatively lower when compared to the costs incurred in RT. The highest value of $R^{P,RT}$ is observed in Scenario P100 G1000, amounting to 3.6 K€/day, followed by Scenario P100 G500 at 2.12 K€/day, and Scenario P50 G1000 at 1.96 K€/day. Additionally, several scenarios show zero $R^{P,RT}$, indicating that water pumps do not always participate in the flexibility service. The findings indicate that changes in availability costs predominantly influence the system's behaviour in RT, aligning with our initial assumptions. That is, availability cost serves as the primary regulatory factor for flexibility dispatch in our model. Moreover, the results suggest that by selecting an optimal RT market participation strategy, specifically through the right availability cost, water operators can potentially reduce their operating costs. This can be observed comparing scenarios P5 G100 and P100 G1000, where the difference in net pumping costs amounts to 4.16 K€/day, underscoring the potential for significant savings.

To obtain further insights into the response of the system to different availability costs, four representative scenarios are selected; namely, P5 G1000, P100 G1000, P100 G500 and P500 G500. Fig. 10 shows the scheduled flexibility reserves and DA tank volumes for the selected scenarios of analysis. Pump flexibility is utilized in all scenarios where the availability cost of pumps is lower than the availability cost of generators. The largest pump flexibility reserves are allocated in scenarios

P5 G1000 followed by P100 G1000. In scenario P100 G500 the pump response becomes limited as the difference between the availability costs of pumps and generators decreases. In scenario P500 G500, water pumps do not provide any flexibility. Results show that when the availability costs of generators and pumps are equal, using flexible generators in RT while optimizing the pump power consumption in DA becomes the cheaper and more cost-effective option. This behaviour can be further understood by observing how the different allocation of flexibility reserves affects the DA tank volume schedule. In particular, when larger flexibility reserves are assigned to water pumps, the tanks are filled and emptied in such a way as to accommodate the scheduled pump flexibility. This means that the volume in the tanks changes in coordination with the pump response in RT. In contrast, when less or even no flexibility reserves are provided by the water pumps, the tanks are filled at times when increasing the nominal (DA) pump load leads to the maximum social welfare and overall system utility.

Fig. 11 shows the effect of the choice of different availability costs on the MCPs. The MCPs for the baseline case (i.e., scenario P5 G500) are shown in red. The mean MCPs obtained by averaging all scenarios are shown in grey. The black error bars indicate the standard deviation. Results show that the choice of availability cost has a relatively low impact on the formation of the MCPs. This is in line with previous observations, such as the small variation in DA system behaviour as seen in Table 2. The most significant change in MCPs is observed at hour 16. At this time, the MCP in the baseline case is 190 €/MWh which is relatively higher than the mean MCP. Other significant changes in MCPs are observed at hours 9, 13 and 21. In general, changes in MCPs can be attributed to different DA schedules of conventional generators and water pumps. For example, generators with a relatively lower marginal production cost may not be fully dispatched in the DA stage as they commit part of their total generation capacity to provide flexibility in RT. Another possible cause is the changing DA pump schedule which, in turn, affects the load on the power system, the dispatch of generators and thus the MCPs. Also, considering that pump bids are set at the maximum auction price in the market, even a small change in DA pump schedule may affect price formation. It must be noted that, although the proposed model and analytical discussions are inherently general, additional simulations should be performed in future studies to evaluate the effect of different system configurations, different market conditions, and levels of congestion in the power network.

By comparing the flexibility dispatched in the four scenarios presented in Fig. 10 with the MCP at the respective hours shown in Fig. 11 (note that the MCPs remain approximately the same in all scenarios), it is observed that under specific conditions there could be a cross-temporal flexibility exchange taking place between flexible generators and water pumps at certain times. For instance, in scenario P5 G1000, between hours 1 and 12 flexible generators increase their power output in RT to a greater extent compared to the actual wind power deviation. At these times, the additional power produced by generators is used by the water pumps. A similar behaviour is observed in scenarios P100 G1000 and P100 G500, although to a lesser extent. As the difference between the availability cost of generators and pumps reduces, the flexibility provided by water pumps becomes increasingly limited. When the availability cost of generators and pumps is equal and larger than the MCP, i.e. in scenario P500 G500, water pumps do not provide flexibility anymore. Results suggest that, under certain availability cost settings, relatively high wind power deviation and MCPs drive the system to reduce pump loads. Hence, water pumps are encouraged to accumulate additional flexibility provided by generators at times when wind power deviation and MCPs are relatively low. This enables the system to reduce pump load even further at times when it is most profitable while preventing the dispatch of more expensive generating units. Although it may seem counter-intuitive, this can help the system to allocate flexibility at the lowest possible price. Results suggest that, based on the availability cost of flexible assets, MCPs,

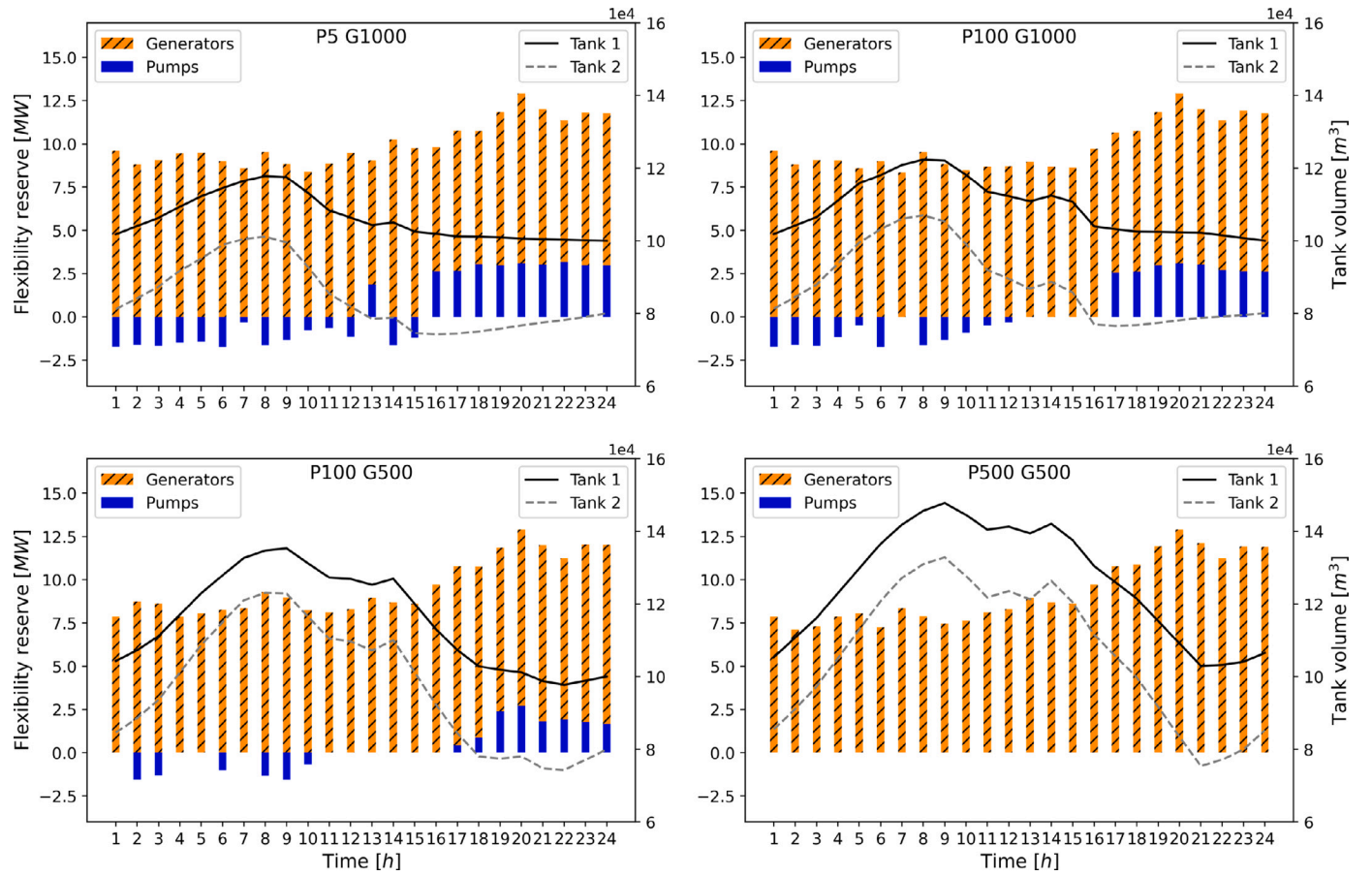


Fig. 10. Sensitivity of flexibility reserves and tank volumes to different availability costs for scenarios P5 G1000 (top left), P100 G1000 (top right), P100 G500 (bottom left) and P500 G500 (bottom right). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

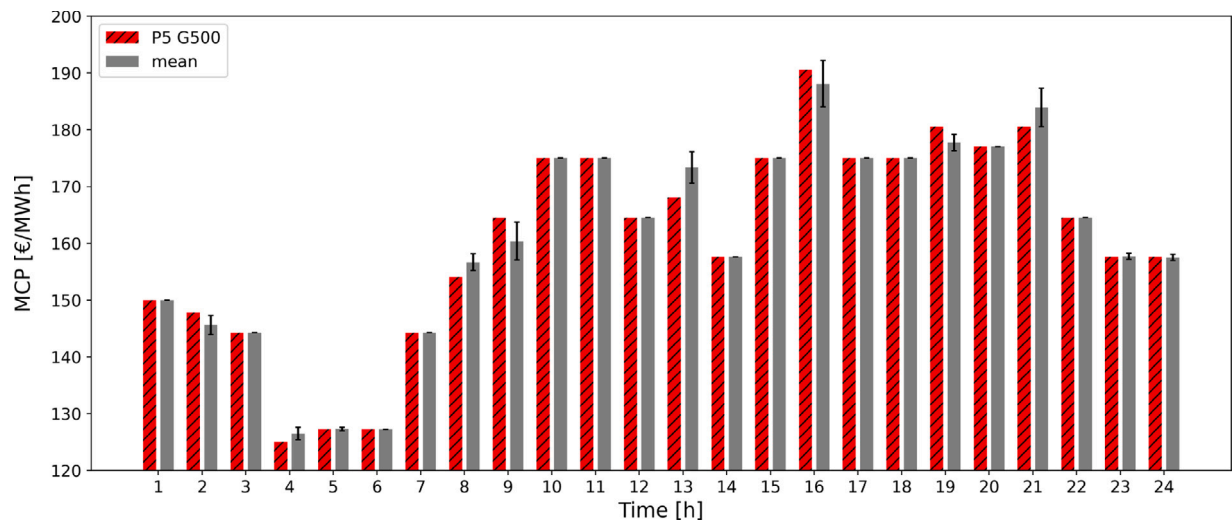


Fig. 11. Sensitivity of MCPs to different availability costs for scenario P5 G500 (red) and the mean of all scenarios (grey); the error bar (black) shows the standard deviation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and wind power deviation, it can be beneficial for some flexible assets to go in the opposite direction to the system needs at certain times (i.e., cross-temporal exchange of flexibility) if it helps to increase the overall system utility. These results show that the smart operation of the coupled power and water system can effectively contribute to increasing the level of flexibility available in the power system.

4.4. Exactness of convex relaxations

This section examines the error introduced by applying the convex relaxation of the friction head losses (13a) and (13c), pump energy consumption (18a) and (18c), and flexibility functions (26a) and (26b) introduced above. In this context, relaxation error is defined as the difference between the left-hand side (lhs) and right-hand side (rhs) of each of the relaxed constraints. Taking constraint (13a) as an example, the mean absolute error (MAE) and normalized root mean square error (NRMSE) are defined as follows

$$MAE = \left[\frac{1}{n(\mathcal{E})n(\mathcal{T})} \left(\sum_{i \in \mathcal{E}} \sum_{t \in \mathcal{T}} |rhs_{i,j,t}^* - lhs_{i,j,t}^*| \right) \right] \quad (27a)$$

$$NRMSE = \left[\frac{1}{n(\mathcal{E})n(\mathcal{T})} \left(\sum_{i \in \mathcal{E}} \sum_{t \in \mathcal{T}} \frac{rhs_{i,j,t}^* - lhs_{i,j,t}^*}{\max(rhs^*) - \min(rhs^*)} \right)^2 \right]^{1/2} \quad (27b)$$

where $n(\mathcal{E})$ and $n(\mathcal{T})$ indicate the number of elements in set $\mathcal{E}$ and set $\mathcal{T}$ respectively. The terms rhs and lhs represent the right-hand side and left-hand side of the constraint. Superscript (*) indicates the value obtained at the optimal solution and $\max(rhs)^* - \min(rhs)^*$ is the difference between the minimum and maximum value of the right-hand side of the constraint. While MAE is an absolute measure of error and has a physical meaning, the NRMSE is a normalized measure of error and it is unitless. The former should be interpreted as the expected average deviation between the right-hand side and the left-hand side of the constraint. The latter is a normalized probabilistic measure expressing the error variance between the left and right-hand side of the constraint. NRMSE should be interpreted as the extent to which the convex relaxation is expected to introduce significant errors. The advantage of NRMSE compared to MAE is that it enables the comparison of relaxation errors between variables of different magnitudes. In this context, if $NRMSE \approx 0$ the convex relaxation can be considered exact. It is worthwhile noting that, a nonzero relaxation error means that it is not possible to give a correct physical interpretation of the results. Table 3 shows the MAE and NRMSE for selected constraints. For $\epsilon = 0.1$, the analysis shows that constraints (18a), (26a) and (26b) are approximately exact, while constraints (13a), (13c) and (18c) are more likely to introduce larger relaxation errors and, therefore, they may be considered inexact. However, it should be noted that the DRCC methodology admits the violation of the chance constraints with a probability level of $\epsilon = 0.1$. This suggests that it is unlikely to observe a zero relaxation error. This is in line with the findings of [10]. While MAE and NRMSE provide an indication of the expected relaxation error introduced by the proposed relaxations, a more detailed analysis of the impact of the choice of the violation probability level ϵ should be conducted in future research.

A heat map is used to obtain further insights into the temporal and spatial distribution of the relaxation errors. Fig. 12 shows a heat map of the relaxation error (i.e., the normalized difference between left and right-hand sides) associated with constraints (13a) and (13c) for each water pipeline at each hour of the day. The results show that the relaxation error for constraint (13a) is particularly significant in pipelines 4, 5, 6 and 9 for all hours. In contrast, the relaxation error associated with constraint (13c) is more significant in pipelines 7 and 10. Further, the temporal distribution of the relaxation error in (13c) is subject to a significant variation between hours 1 and 15 with particular peaks at hours 7, 11 and 15. The results suggest that the nonlinear dynamics of the water network, the affine response by water pumps, and the indirect control of the water storage tanks

Table 3

Calculated mean absolute error (MAE) and normalized root mean square error (NRMSE).

Constraint	MAE	NRMSE
(13a)	$1.87 \cdot 10^1$	$2.33 \cdot 10^{-1}$
(13c)	$4.57 \cdot 10^{-2}$	$1.44 \cdot 10^{-1}$
(18a)	$1.47 \cdot 10^{-3}$	$1.27 \cdot 10^{-3}$
(18c)	$2.06 \cdot 10^{-2}$	$3.44 \cdot 10^{-1}$
(26a)	$2.70 \cdot 10^{-3}$	$4.07 \cdot 10^{-3}$
(26b)	$1.16 \cdot 10^{-2}$	$5.22 \cdot 10^{-2}$

and pressure-reducing valves contribute to these errors. This is in line with relevant literature on this topic. As discussed in [29], the convex hull relaxation of the friction head losses is likely to introduce errors. Further, the McCormick relaxation for the product of two continuous variables is inexact [49]. Other sources of errors are attributable to the SOC relaxation of probabilistic chance constraints and the inherent violation probability level ϵ . Possible enhancements to overcome these issues will be considered in future work.

5. Conclusion

This paper introduces a novel market-based mechanism that allows for short-term scheduling and operation of integrated power and water networks. The problem is formulated as a distribution robust chance-constrained (DRCC) optimization problem. It maximizes utility in the power sector by using flexibility in both the power and water sectors to compensate for possible imbalances. The imbalances are assumed to occur due to imperfect wind power forecasts. The way the DRCC optimization problem is formulated makes it possible to study (1) the economic advantages of utilizing the flexibility that is embedded in the water network, and (2) the impacts of uncertainty associated with wind power deviations on the operation of a coupled power and water network. Here we assumed decisions about flexibility allocation are made in a centralized and fully competitive market set-up. Under the assumption of affine response to uncertainty, the proposed mechanism determines the optimal flexibility dispatch such that it maximizes DA social welfare. At the same time, it ensures that the imbalances resulting from the worst-case wind deviations are compensated for in RT. Two sources of flexibility are used: (1) flexible generators and (2) water pump loads.

A novel, hybrid, convex hull-based relaxation technique is proposed to solve the DRCC problem. The proposed relaxation enables overcoming the non-linearity of the water network thus making the DRCC problem computationally tractable. The proposed coordination mechanism enables the utilization of flexibility embedded in the water network by adjusting the operation of water pumps connected to water storage tanks in RT. This results in a resilient, network-aware, and regret-free solution for the DA scheduling of power and water networks. The relaxation error is observed to be small but not always negligible. In particular, the approximation of friction head losses introduces relatively higher relaxation errors compared to other constraints and is shown to be the main source of inaccuracy in the model.

Under the assumptions made, we show that the power system can exploit the existing flexibility in the water network. The coordinated operation of elevated water storage tanks and water pumps acting as demand-response providers can effectively compensate for power system imbalances, covering up to 25% of the total wind power mismatch. Both sectors benefit financially from dispatching flexibility as determined by the proposed mechanism. The benefits for the power sector are lower costs of flexibility procurement. The benefits for the water sector are lower overall energy costs, quantified in thousands of euros per day. Sourcing flexibility from the water system ensures higher

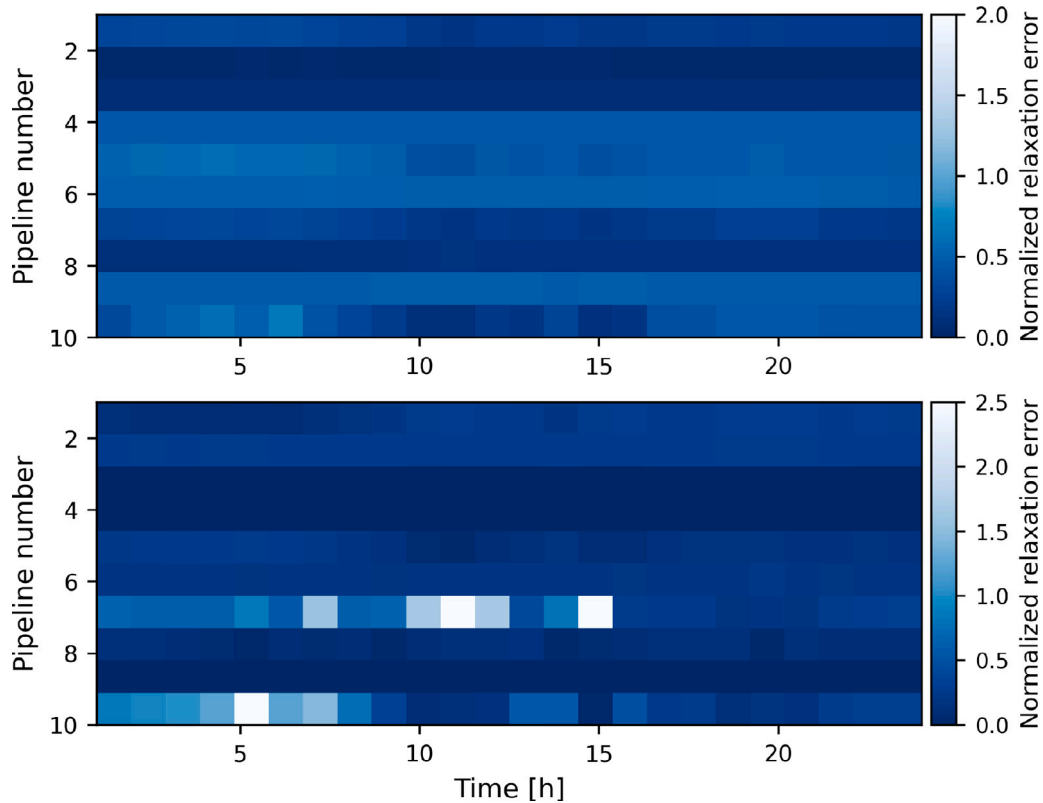


Fig. 12. Normalized relaxation error for constraint 13a (top) and 13c (bottom) for each water pipeline at each hour of the day. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

reliability in the power system, with no impact on the water system end-users. This supports the hypothesis that integrated, market-based coordination of power and water networks can potentially lead to a higher overall utility and system-wide resiliency.

5.1. Limitations and future work

The research has specific limitations. Firstly, our focus on power and water connections implies a one-way energy flow, limiting the potential flexibility exchange between interconnected networks. Secondly, our uncertainty modelling approach, based on data-driven first- and second-order moments, depends heavily on data accuracy and availability. Lastly, the inherent errors introduced by convex relaxations alter the physical interpretability of the results.

In the future, we will extend our study to include bilateral power-water network connections, by integrating power-producing technologies such as water turbines. Further, we will explore methods for refining the convex relaxation of water network constraints, such as the tightening of McCormick relaxations. Comparative analysis with other DRCC modelling methods, such as distance-based approaches, will also be explored.

Another area of future research is to consider a broader range of uncertainties, including solar generation, as well as power and water demands. Additionally, we aim to investigate the impact of strategic behaviour among flexible assets, such as generators and pumps, on market outcomes, flexibility dispatch programs, and procurement costs. Lastly,

our scope will extend to studying uncertainty-aware infrastructure investment planning within integrated water and power systems.

CRediT authorship contribution statement

A. Belmondo Bianchi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **J. Willet:** Conceptualization, Methodology, Software, Validation, Resources, Data curation, Writing – review & editing, Visualization, Supervision, Project administration. **H.H.M. Rijnaarts:** Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition. **S. Shariat Torbaghan:** Conceptualization, Methodology, Software, Resources, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Annex

See Tables A.1–A.5.

Table A.1Bus ID, load ID, distribution of total system demand, and estimated bids of conventional loads U^D and pumps U^P .

Bus ID	Load ID	Demand multiplier	U^D [€/MWh]	U^P [€/MWh]
1	1	0.038	200	–
2	2	0.034	250	–
3	3	0.063	200	–
4	4	0.026	150	–
5	5	0.025	225	–
6	6	0.048	125	–
7	7	0.044	250	–
8	8	0.06	300	–
9	9	0.061	150	–
10	10	0.068	250	–
11	P1	–	–	350
12	P3	–	–	350
13	11	0.093	225	–
14	12	0.068	175	–
15	13	0.111	300	–
16	14	0.035	350	–
17	–	–	–	–
18	15	0.117	250	–
19	16	0.064	200	–
20	17	0.045	250	–
21	P2	–	–	350
22	P4	–	–	350
23	–	–	–	–
24	–	–	–	–

Table A.2Generator ID, Bus ID, upper and lower generation limits ($\bar{P}^G$, $\underline{P}^G$), and marginal production cost of generators (C^G).

Generator ID	Bus ID	Type	$\bar{P}^G$	$\underline{P}^G$	C^G
1	1	conventional	15.2	0	200
2	2	conventional	15.2	0	175
3	7	conventional	30	0	328.05
4	13	conventional	59.1	0	154.1
5	15	conventional	6	0	110
6	15	conventional	15.5	0	115
7	16	conventional	15.5	0	220
8	18	conventional	40	0	104.2
9	21	conventional	40	0	134.5
10	22	conventional	30	0	120
11	23	conventional	6	0	105
12	23	conventional	35	0	161.1
13	5	wind	50	0	0
14	7	wind	50	0	0

Table A.3Line ID, connection from bus to bus, reactance, and line capacity ($\bar{f}$).

LineNum	From	To	Reactance [p.u.]	$\bar{f}$ [MVA]
1	1	2	0.0146	17.5
2	1	3	0.2253	17.5
3	1	5	0.0907	50
4	2	4	0.1356	17.5
5	2	6	0.205	17.5
6	3	9	0.1271	17.5
7	3	24	0.084	40
8	4	9	0.111	17.5
9	5	10	0.094	50
10	6	10	0.0642	17.5

(continued on next page)

Table A.3 (continued).

LineNum	From	To	Reactance [p.u.]	$\bar{f}$ [MVA]
11	7	8	0.0652	100
12	8	9	0.1762	22.5
13	8	10	0.1762	17.5
14	9	11	0.084	40
15	9	12	0.084	40
16	10	11	0.084	40
17	10	12	0.084	40
18	11	13	0.0488	50
19	11	14	0.0426	50
20	12	13	0.0488	50
21	12	23	0.0985	50
22	13	23	0.0884	50
23	14	16	0.0594	50
24	15	16	0.0172	50
25	15	21	0.0249	50
26	15	24	0.0529	50
27	16	17	0.0263	50
28	16	19	0.0234	50
29	17	18	0.0143	50
30	17	22	0.1069	50
31	18	21	0.0132	100
32	19	20	0.0203	100
33	20	23	0.0112	100
34	21	22	0.0692	50

Table A.4

Node ID, node type, elevation, demand, minimum, maximum, and initial volume (for storage tanks only).

Node ID	Type	Elevation [m]	Demand [m ³ /h]	$\underline{V}$ [m ³]	$\bar{V}$ [m ³]	V^0 [m ³]
1	junction	0	–	–	–	–
2	junction	0	–	–	–	–
3	junction	0	–	–	–	–
4	junction	0	5000	–	–	–
5	junction	0	7000	–	–	–
6	junction	10	–	–	–	–
7	junction	0	4000	–	–	–
8	junction	0	6000	–	–	–
9	source	10	–	–	–	–
10	source	10	–	–	–	–
11	tank inlet	200	–	–	–	–
12	tank	30	–	0	$1.70 \cdot 10^5$	$1.00 \cdot 10^5$
13	junction	0	–	–	–	–
14	junction	0	–	–	–	–
15	junction	5	5000	–	–	–
16	junction	15	7000	–	–	–
17	source	10	–	–	–	–
18	source	10	–	–	–	–
19	tank inlet	200	–	–	–	–
20	tank	30	–	0	$1.36 \cdot 10^5$	$8.00 \cdot 10^4$

Table A.5

Line ID, type, connection from node to node, length, diameter, minimum and maximum capacity.

Line ID	Type	From	To	Length [km]	Diameter [m]	$\underline{Q}$ [m ³ /h]	$\overline{Q}$ [m ³ /h]
1	pipe	1	6	100	2	0	$2.26 \cdot 10^4$
2	valve	2	1	0	2	0	$2.26 \cdot 10^4$
3	pipe	2	3	100	2	0	$2.26 \cdot 10^4$
4	pipe	3	4	100	2	0	$2.26 \cdot 10^4$
5	valve	3	8	0	2	0	$2.26 \cdot 10^4$
6	pipe	4	5	100	2	0	$2.26 \cdot 10^4$
7	pipe	6	7	100	2	0	$2.26 \cdot 10^4$
8	valve	6	15	0	2	0	$2.26 \cdot 10^4$
9	pipe	7	8	100	2	0	$2.26 \cdot 10^4$
10	pump	9	2	0	–	$2.00 \cdot 10^3$	$2.00 \cdot 10^4$
11	pump	10	11	0	–	$2.00 \cdot 10^3$	$2.00 \cdot 10^4$
12	virtual	11	12	0	2	–	–
13	pipe	12	1	100	2	0	$2.26 \cdot 10^4$
14	valve	13	8	0	2	0	$2.26 \cdot 10^4$
15	valve	13	14	0	2	0	$2.26 \cdot 10^4$
16	valve	14	7	0	2	0	$2.26 \cdot 10^4$
17	pipe	14	15	100	2	0	$2.26 \cdot 10^4$
18	pipe	15	16	100	2	0	$2.26 \cdot 10^4$
19	pump	17	14	0	–	$2.00 \cdot 10^3$	$2.00 \cdot 10^4$
20	pump	18	19	0	–	$2.00 \cdot 10^3$	$2.00 \cdot 10^4$
21	virtual	19	20	0	2	–	–
22	pipe	20	13	100	2	0	$2.26 \cdot 10^4$

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